

File System Analysis Using Autopsy and The Sleuth Kit

Course:

Digital Forensics

Instructor:

Prof. Naveed Naeem abbas

Performed By:

Ahmad Talha

Introduction

This lab focuses on understanding hard disks and file systems from a digital forensics perspective. The practical work involves analyzing forensic disk images to identify file system types, examine file and folder contents, extract metadata, and recover files.
The lab is divided into two main parts:

- Analysis of a Linux forensic image using **Autopsy**
- Analysis of Windows forensic images using **The Sleuth Kit (TSK)**



Autopsy

Introduction

Autopsy is a graphical digital forensics platform used to investigate digital evidence from forensic disk images and storage media. It provides a user-friendly interface through which investigators can examine file systems, files, directories, metadata, deleted data, and other digital artifacts.

Autopsy is built on top of various forensic analysis capabilities and allows investigators to organize their examination into structured forensic cases.

Purpose of Autopsy

The main purpose of Autopsy is to make the examination of digital evidence easier through a graphical interface.

In this lab, Autopsy was used to:

- Create and manage forensic cases.
- Add forensic disk images as evidence sources.
- Analyze Linux file systems.
- Examine files and directories.
- View file contents and extracted text.
- Examine file metadata and data artifacts.

- Identify and examine deleted files.
- Analyze different file types such as JPG, TXT, DOC, XLS, DAT, and other files.
- Examine suspicious items.
- Perform keyword-based searches.
- Review evidence in an organized manner.

Advantages of Autopsy

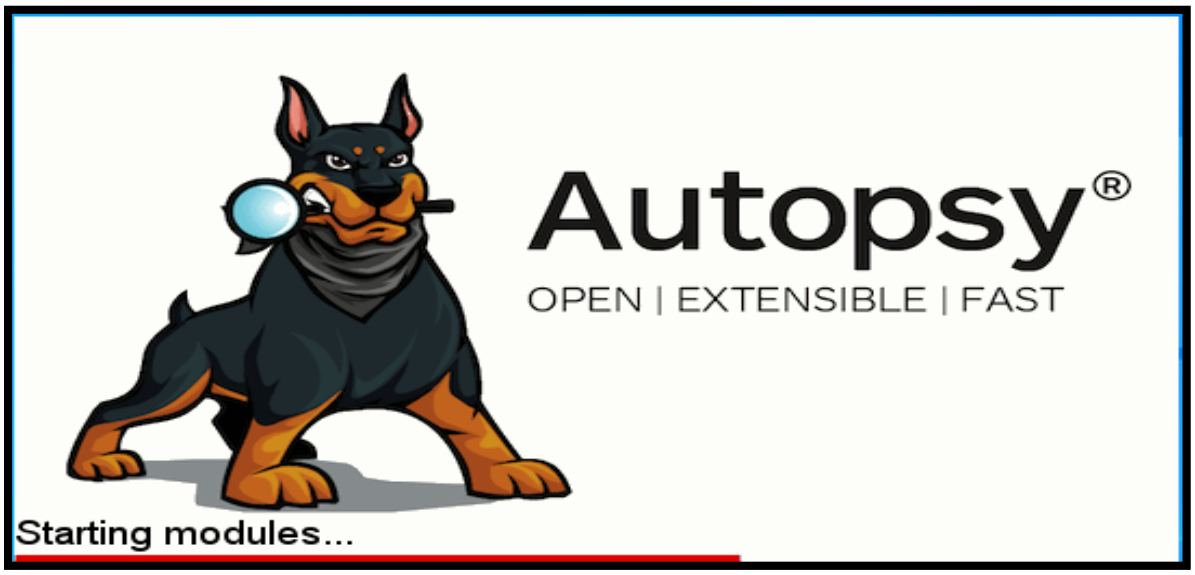
- Provides a **graphical and user-friendly interface**.
- Organizes evidence through forensic cases.
- Supports analysis of different file systems and file types.
- Makes deleted-file analysis easier.
- Provides access to file metadata and extracted content.
- Helps investigators examine large amounts of evidence systematically.
- Provides multiple analysis features within a single forensic platform.

Linux File System Analysis Using Autopsy

Practical Demonstration

01.Installing and Launching Autopsy

Autopsy was installed on the Windows 10 virtual machine. After installation, the application was launched to begin the forensic investigation.



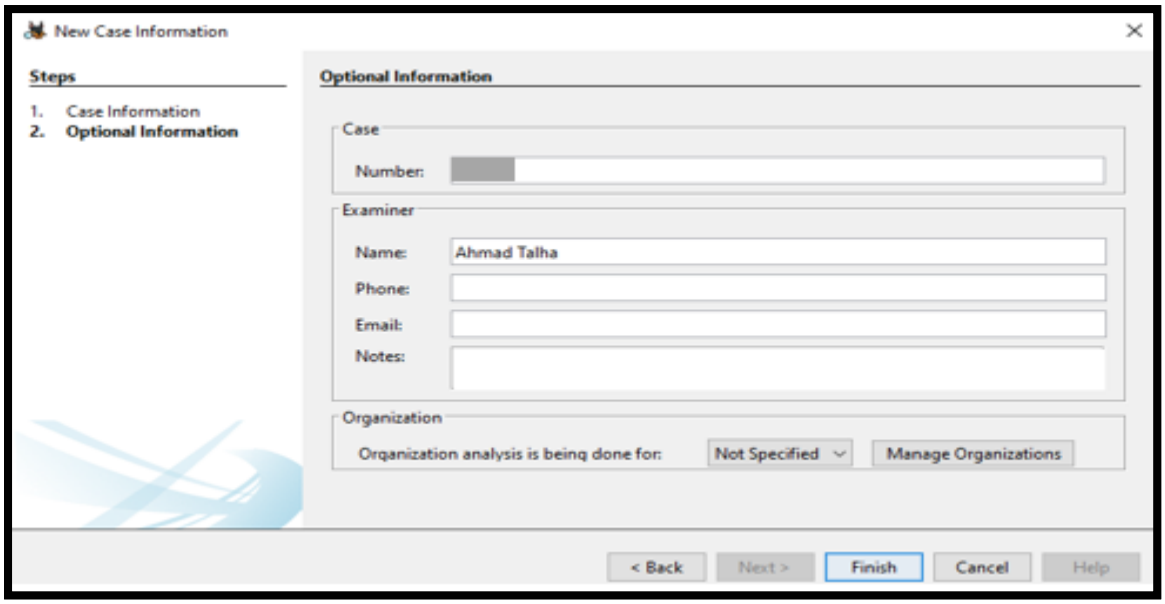
02.Creating a New Case

After launching Autopsy, the **New Case** option was selected to create a new forensic investigation.



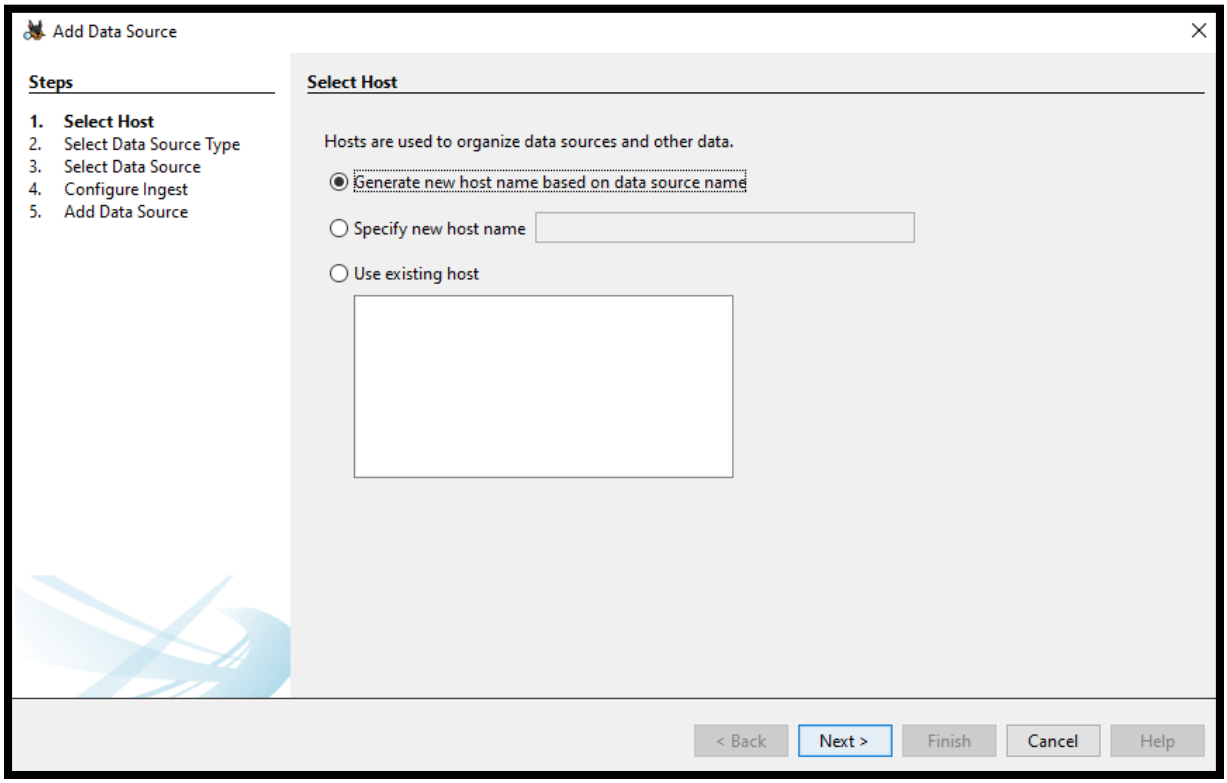
03.Entering Case Information

The required case information was entered to create and organize the forensic investigation. Optional case information was also provided where required.



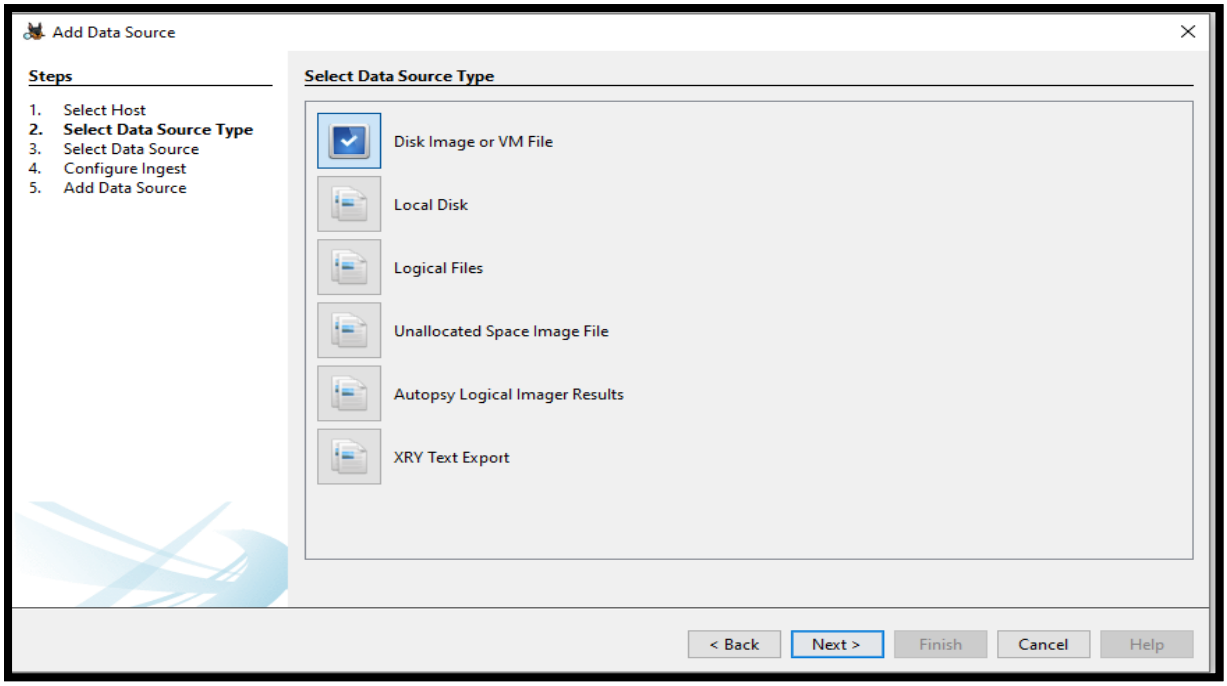
04.Selecting the Host

The appropriate host was selected for the forensic case before adding the evidence source.



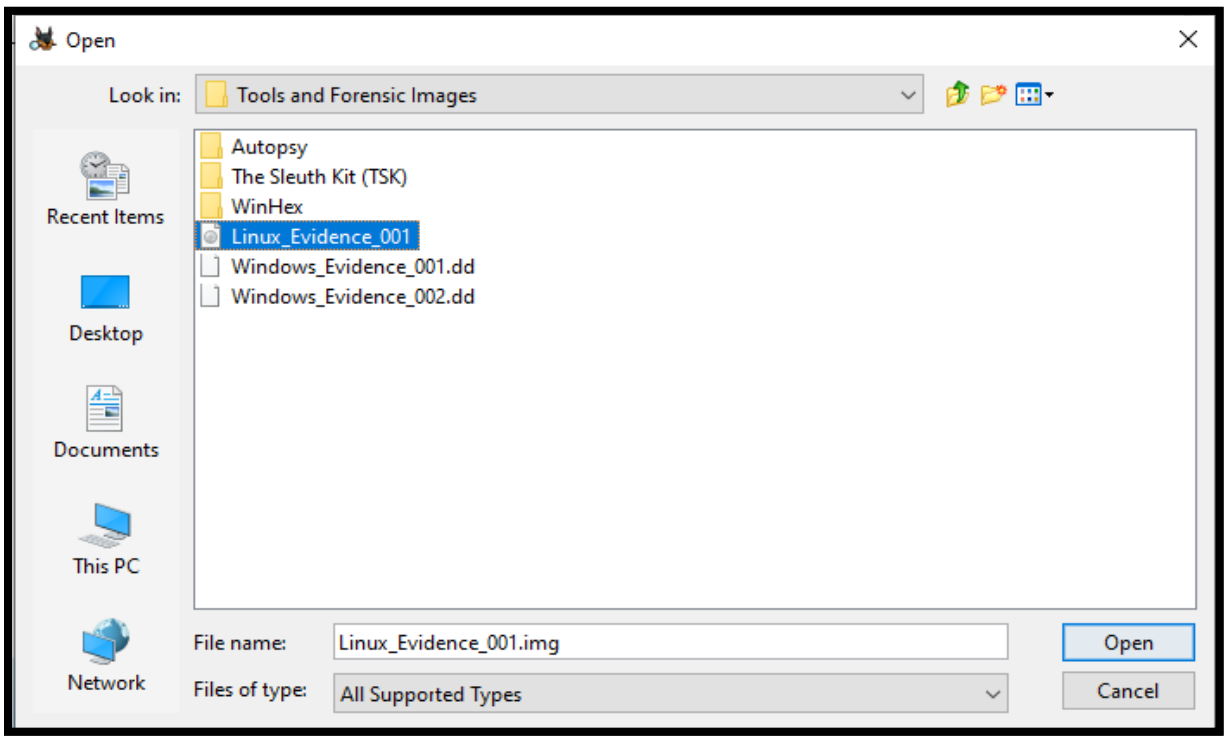
05.Selecting the Data Source Type

The **Disk Image or VM File** option was selected as the data source type because the evidence was provided in the form of a forensic disk image.



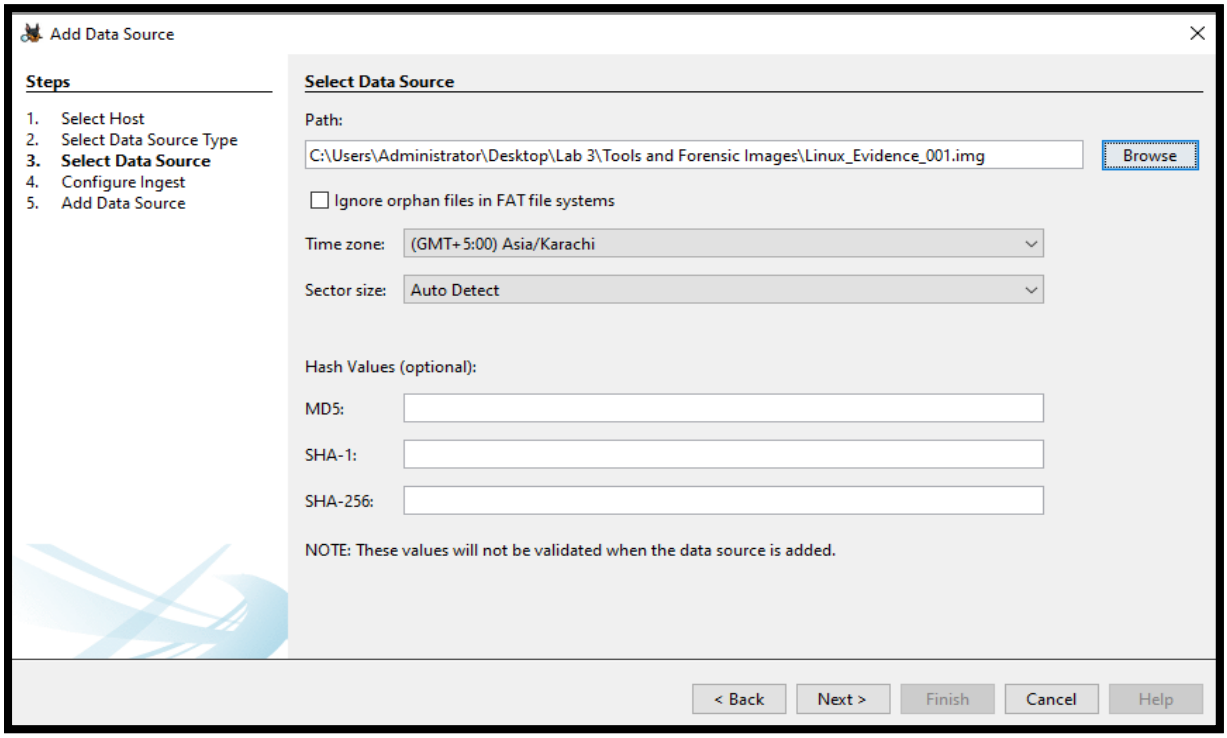
06.Selecting the Forensic Image

The Linux_Evidence_001 image was selected as the forensic evidence source for analysis.



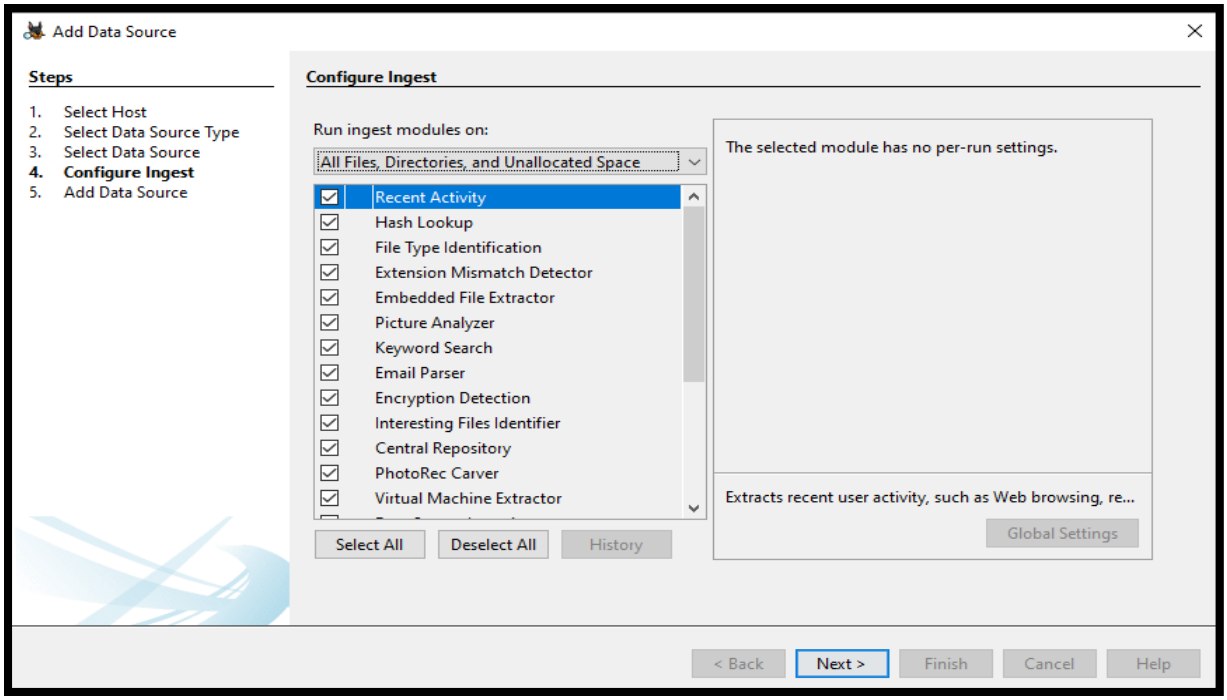
07.Specifying the Data Source Path

The location of the selected forensic image was specified in the Data Source section.



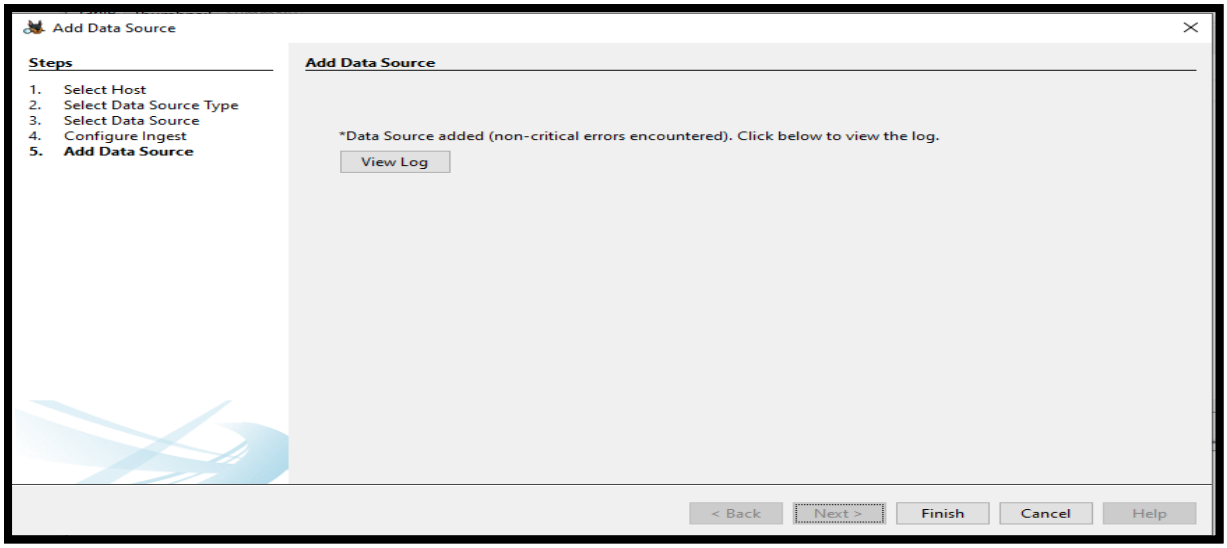
08.Configuring Ingest Modules

The default ingest configuration was retained and the process was continued to the next step.



09.Adding the Data Source

The forensic image was successfully added to the case and became available for analysis in Autopsy.

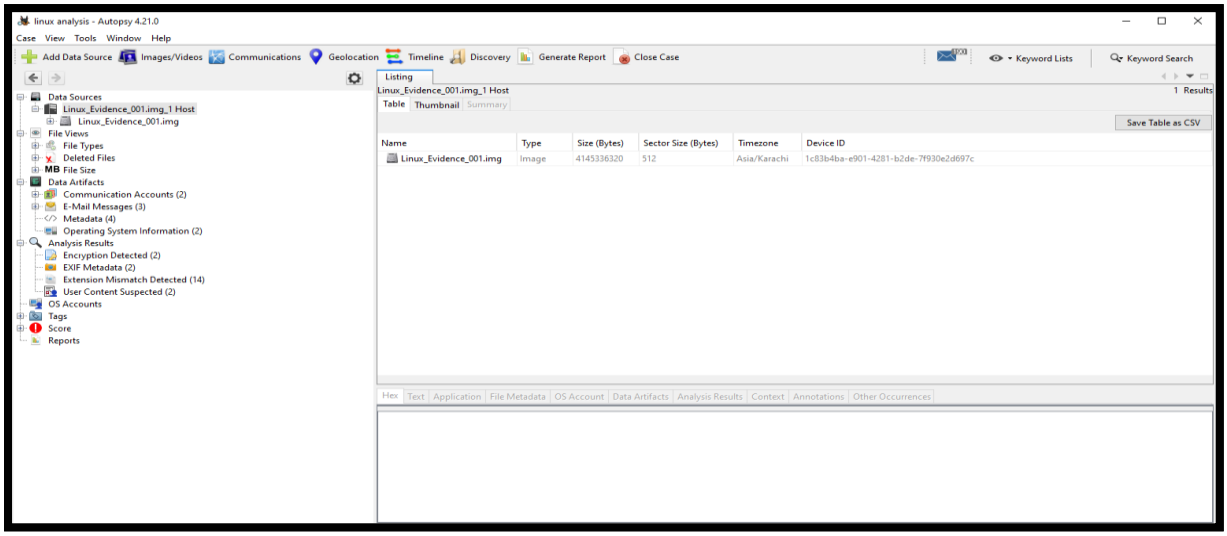


10.Analyzing the Linux File System

After adding the image, Autopsy analyzed the contents of the Linux forensic image. The file system could then be explored through the Autopsy interface.

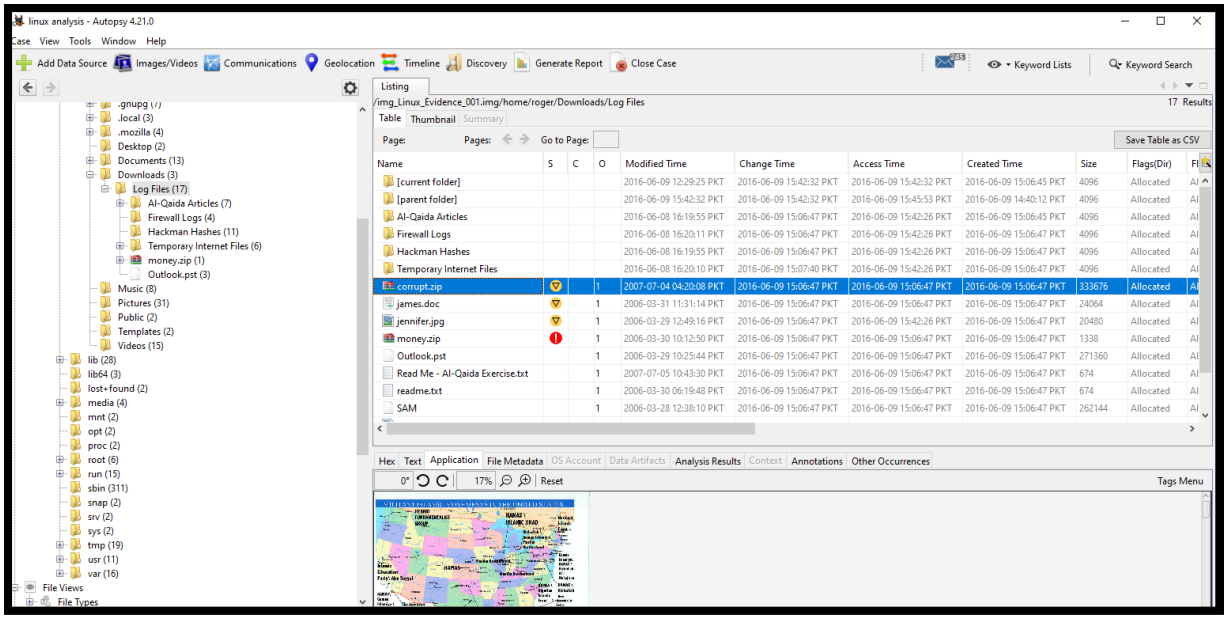
Analysis included:

- Examining directories
- Viewing files
- Navigating through the Linux file system
- Examining specific folders
- Reviewing file contents



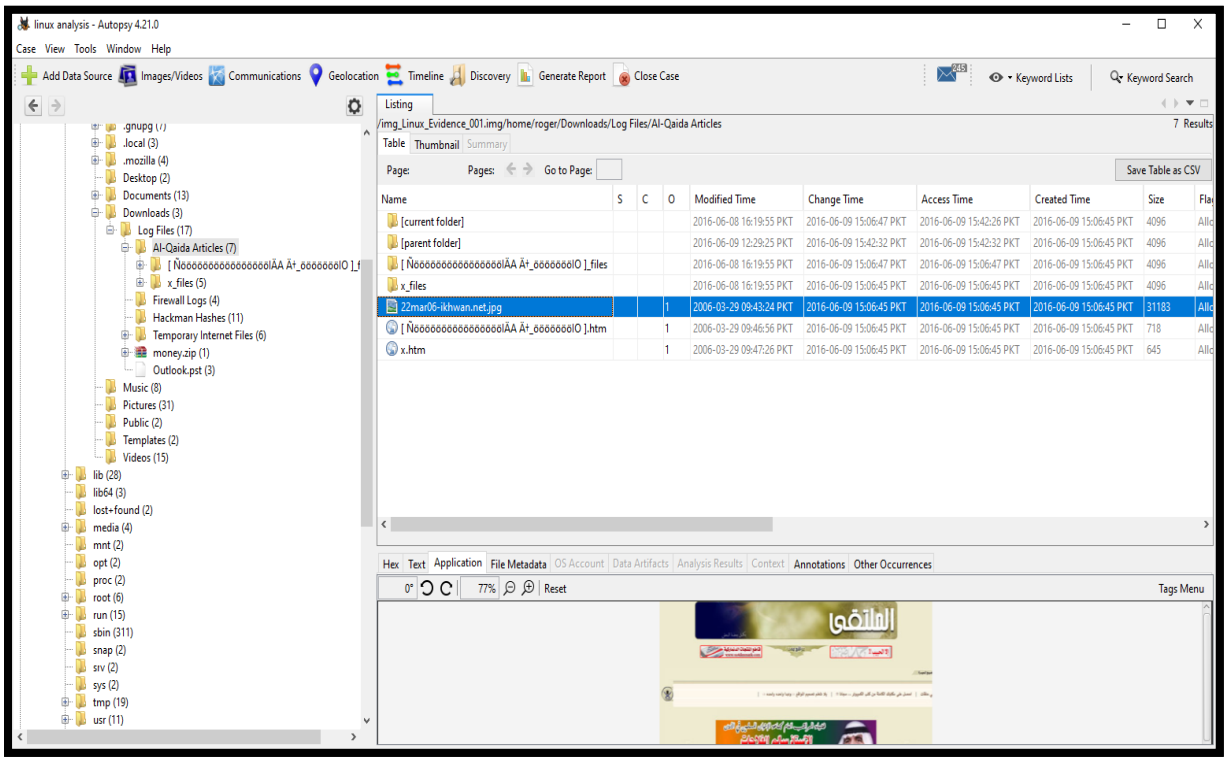
11.Examining Log Files

The **Log Files** section was accessed from the forensic image to examine the files available within the selected category.



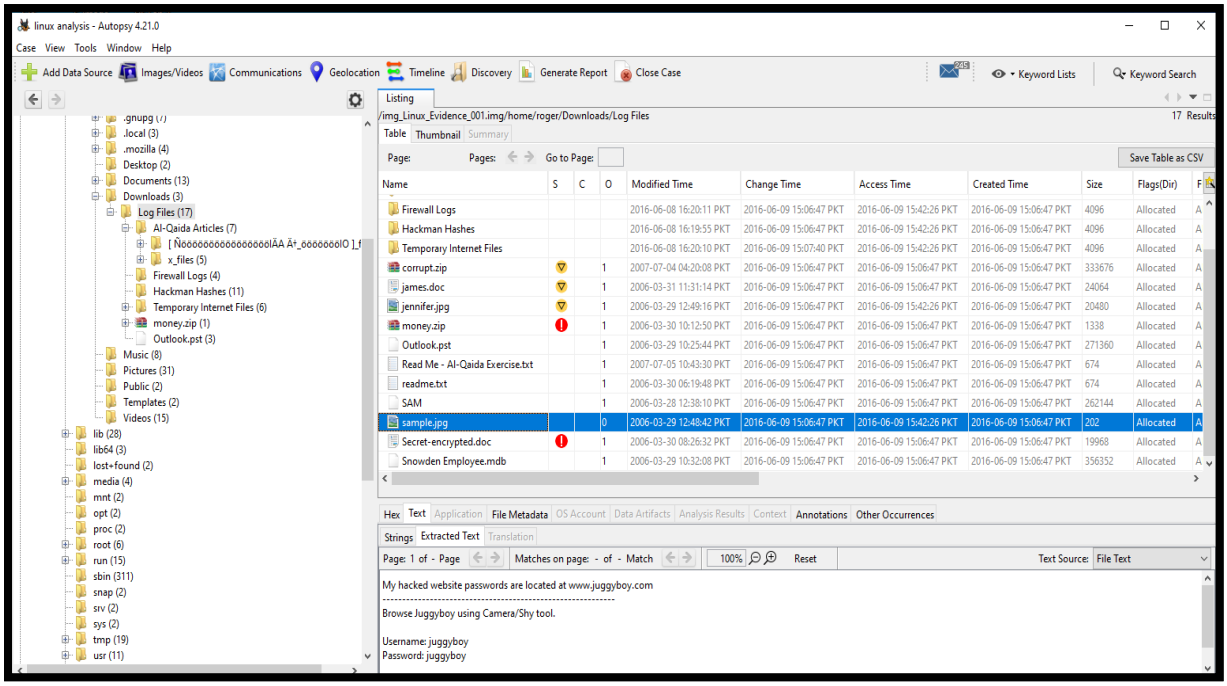
12.Selecting a JPG File

A JPG file was selected from the **Al-Qaida Articles** section for further forensic examination.



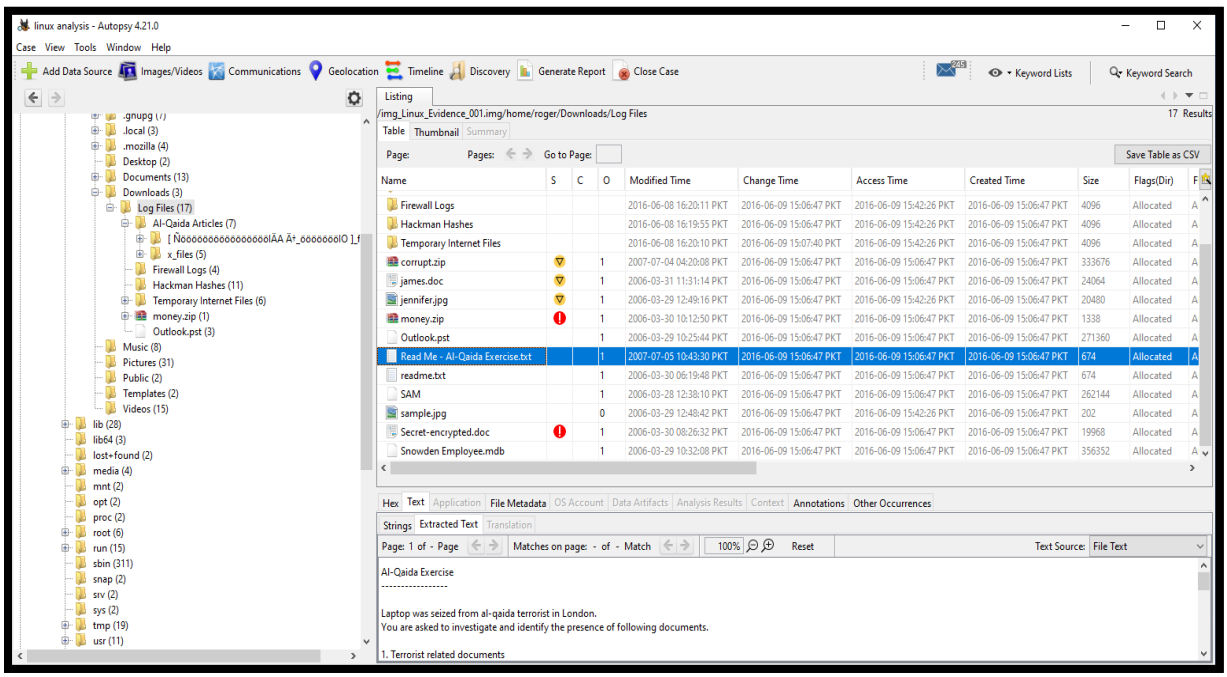
13.Examining Extracted Text from the JPG File

The selected JPG file was examined using the extracted text view to review the text identified from the file.



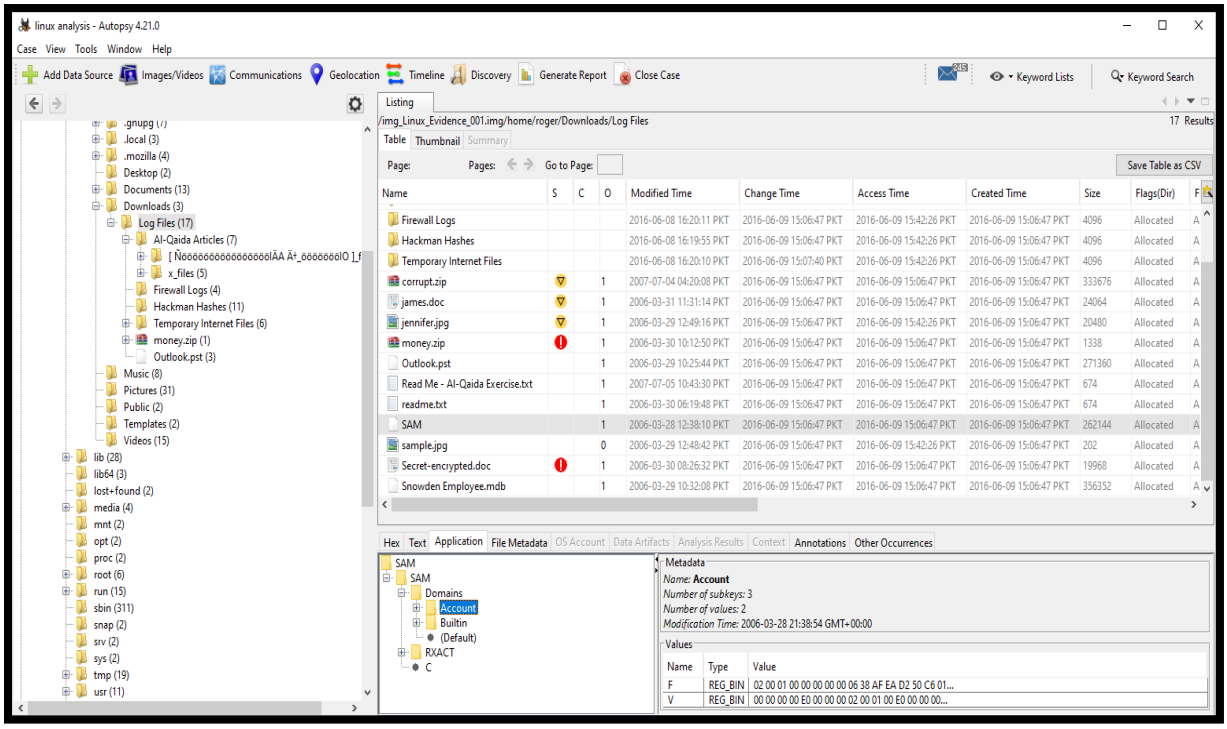
14.Examining a TXT File

A TXT file was selected and its extracted text was examined through the Autopsy interface.



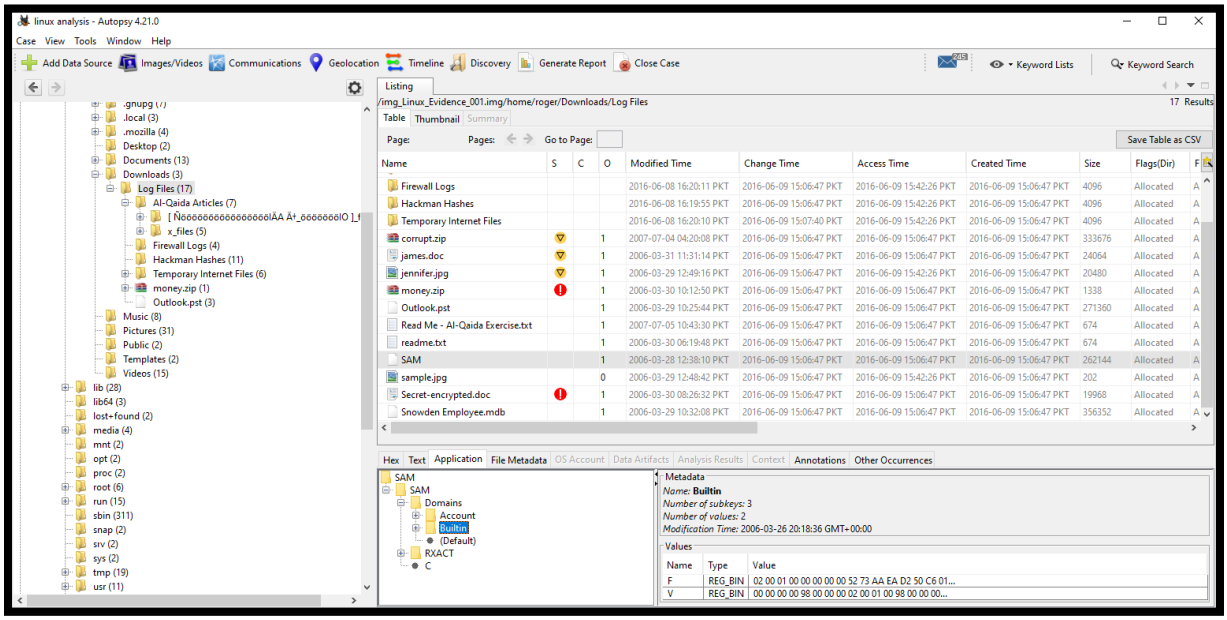
15.Examining the SAM Folder

The **SAM** folder was accessed to examine its contents and the subfolders or related entries available within it.



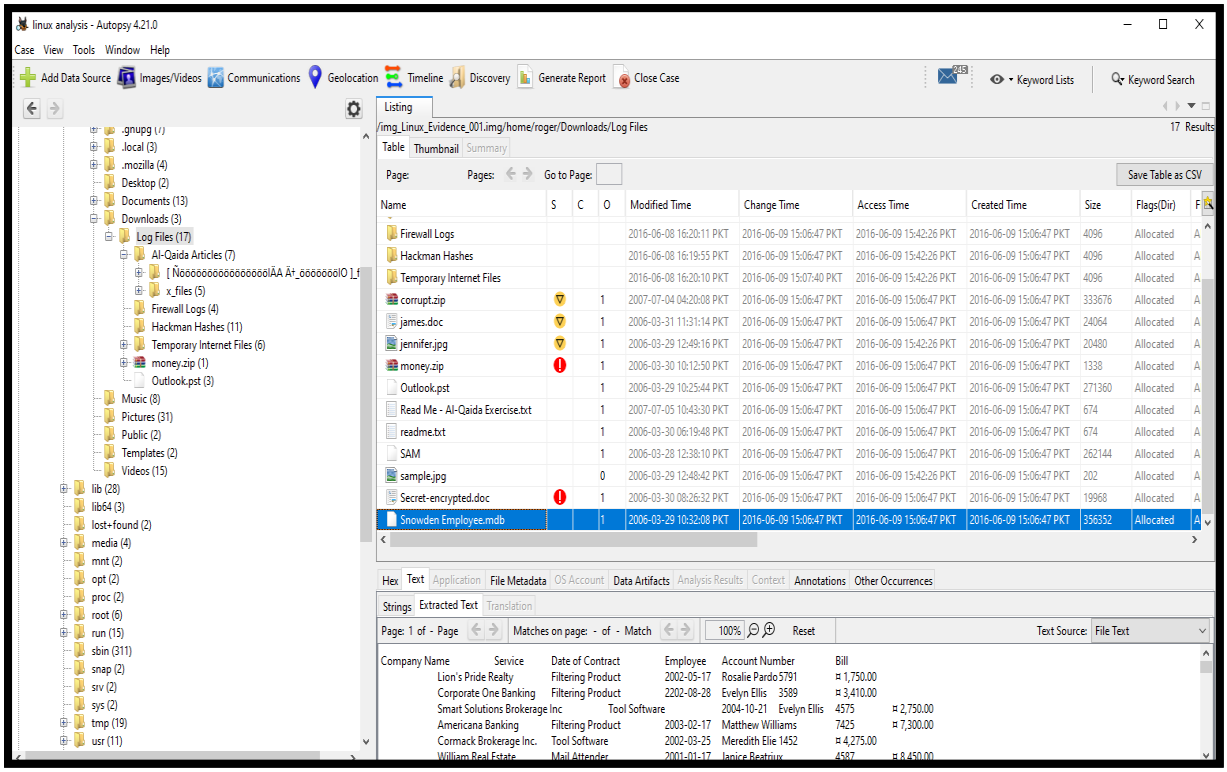
16.Examining File Metadata

The metadata associated with a selected folder or file was examined to identify the available forensic information.



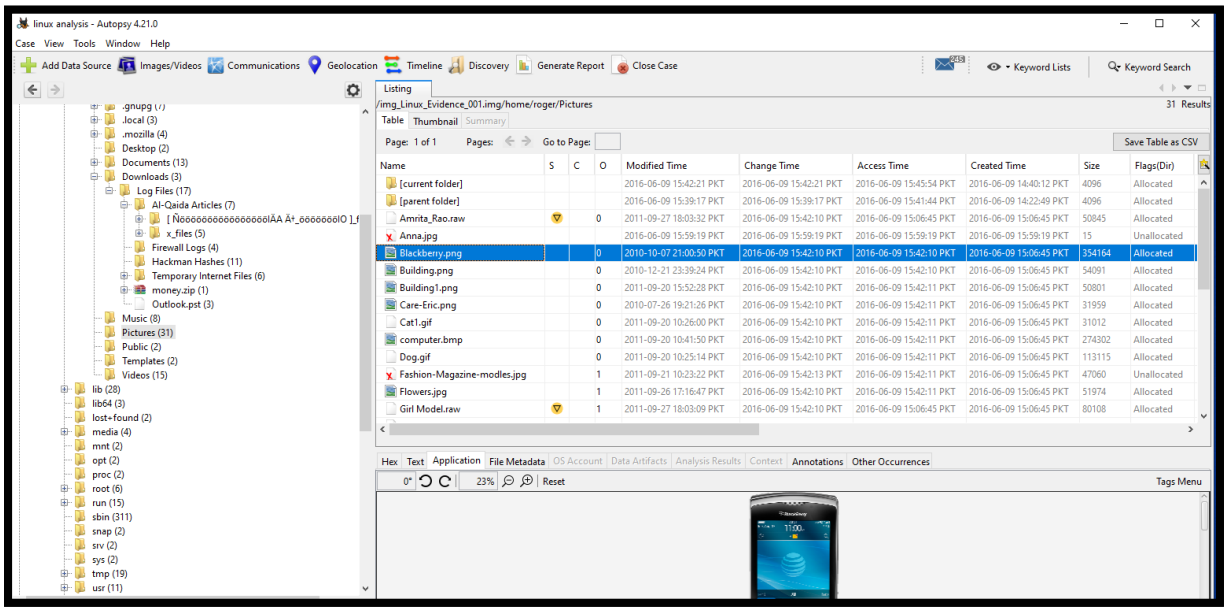
17.Examining an MDB File

An MDB file was selected for examination, and its extracted content was reviewed through Autopsy.



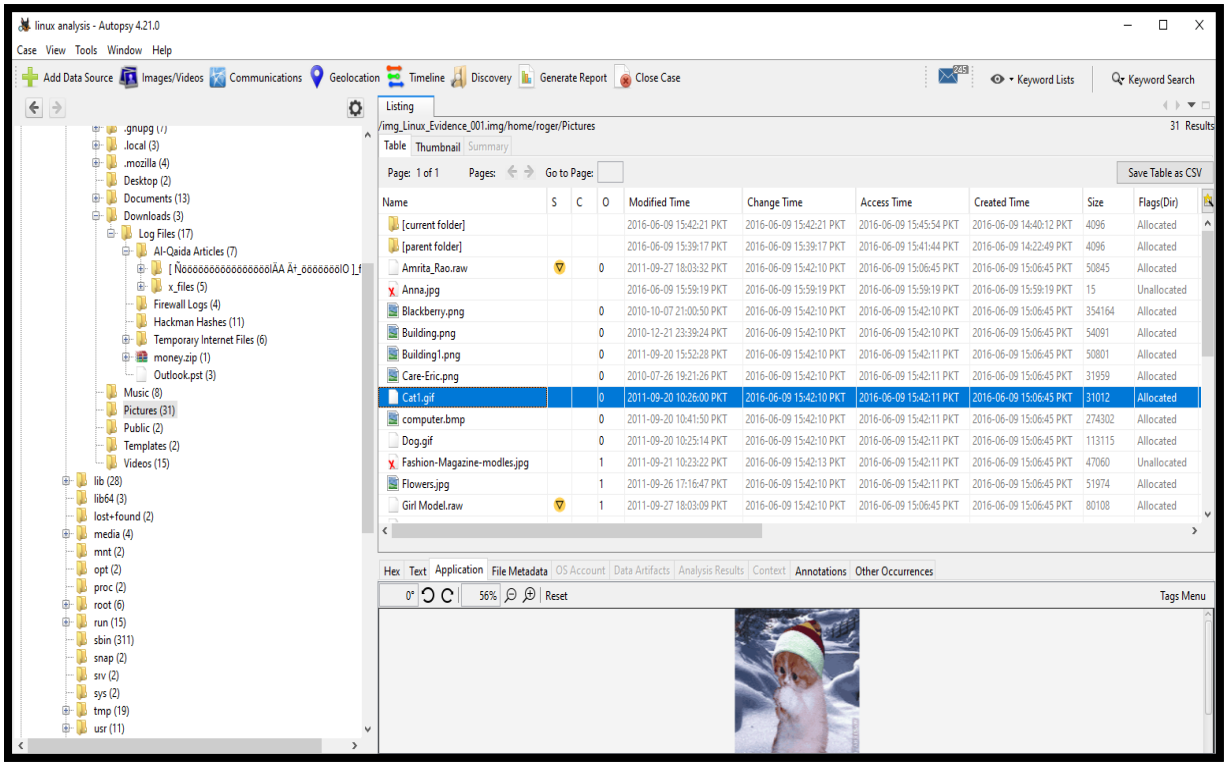
18.Examining an Image File

An image file was selected from the forensic image and examined using Autopsy's analysis interface.



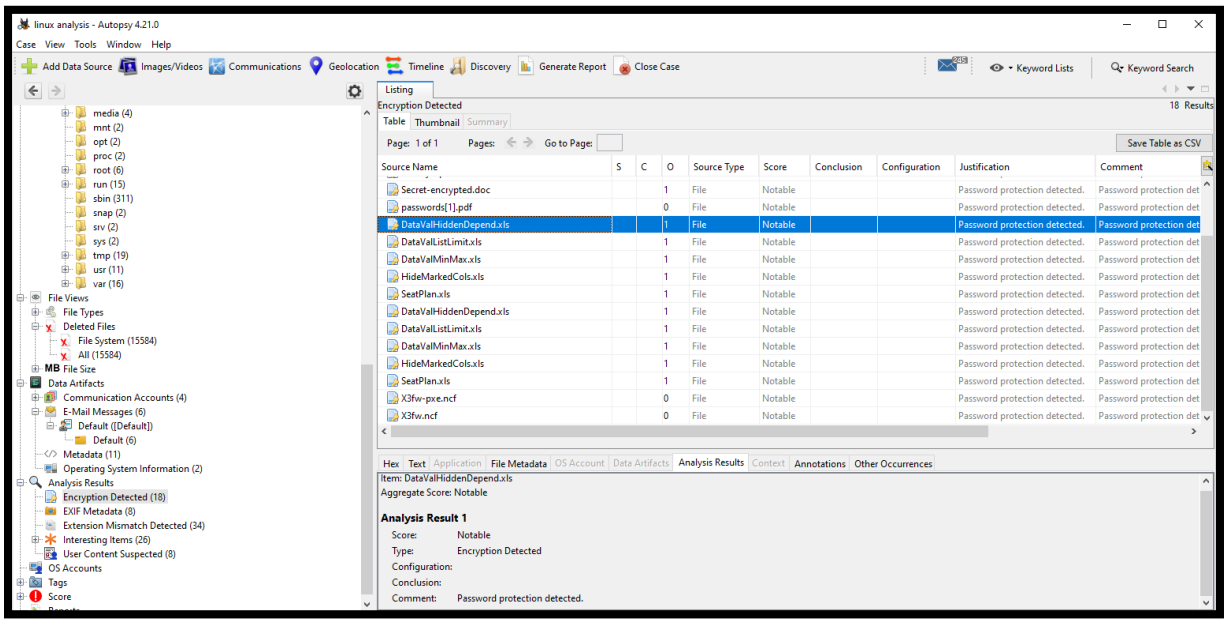
19.Examining Another Image File

Another image file was selected to further examine the available digital evidence and its contents.



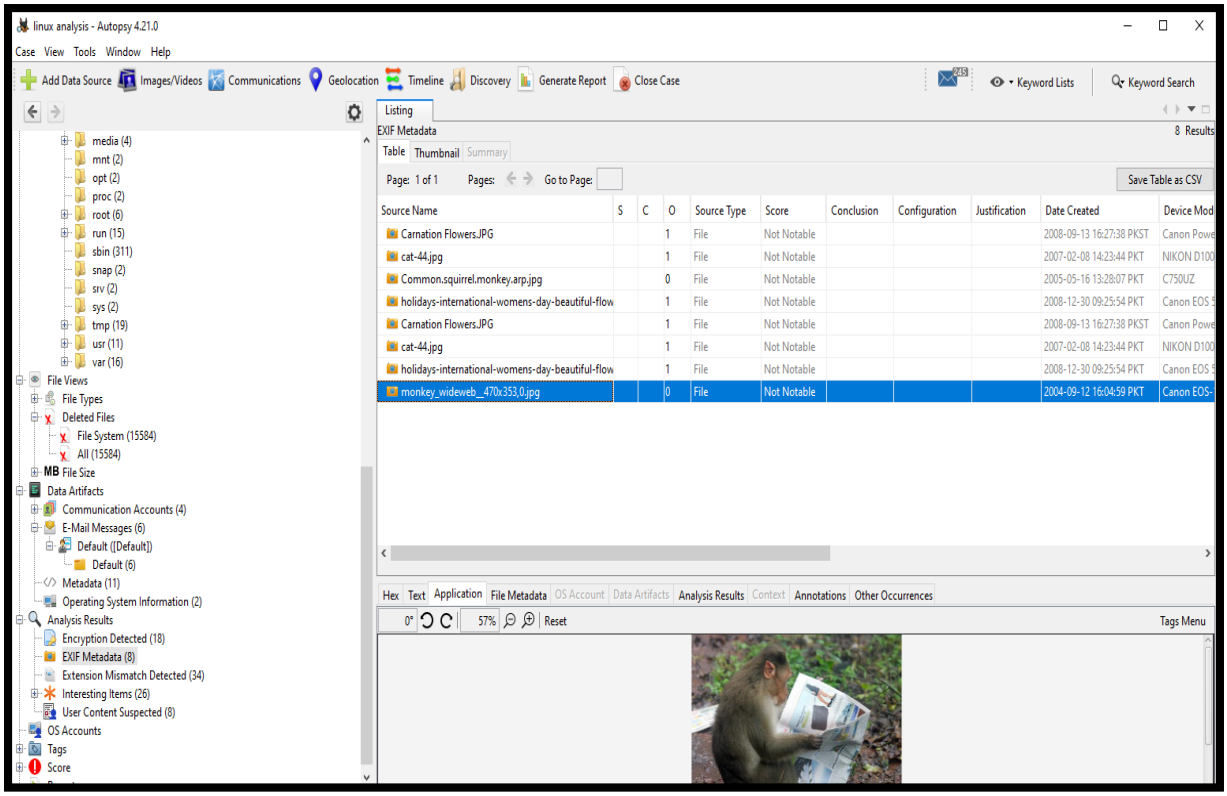
20.Analyzing an XLS File

The XLS file was selected and its available forensic analysis results were examined.



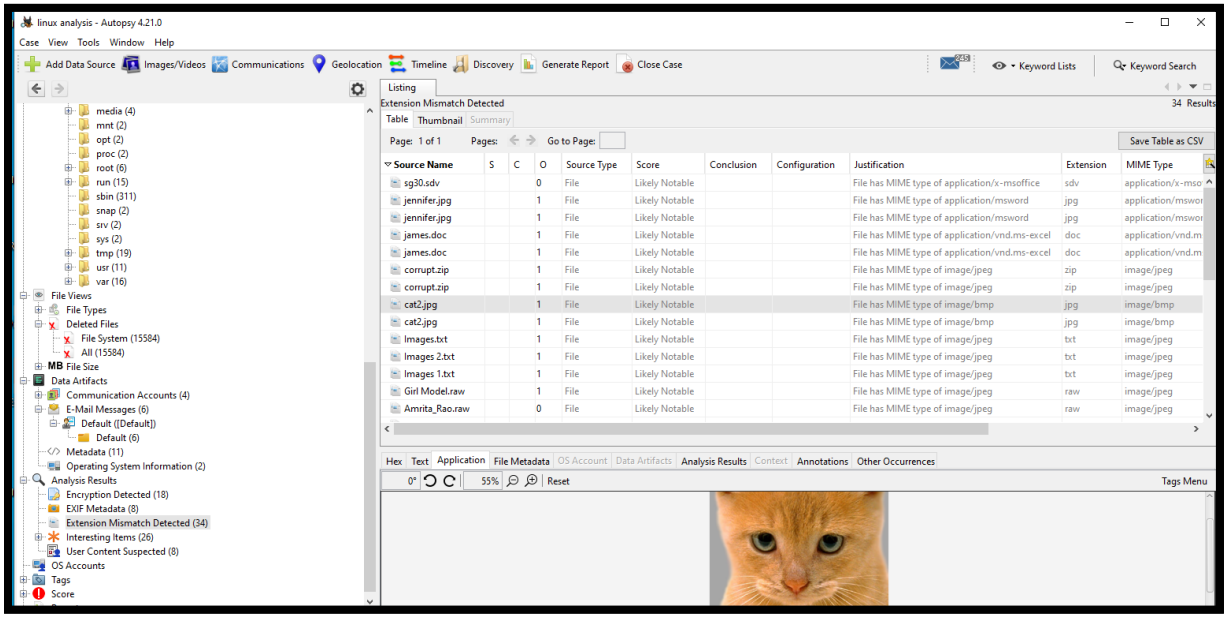
21.Examining an Additional Image File

An additional image file was selected and examined to review its available contents and analysis results.



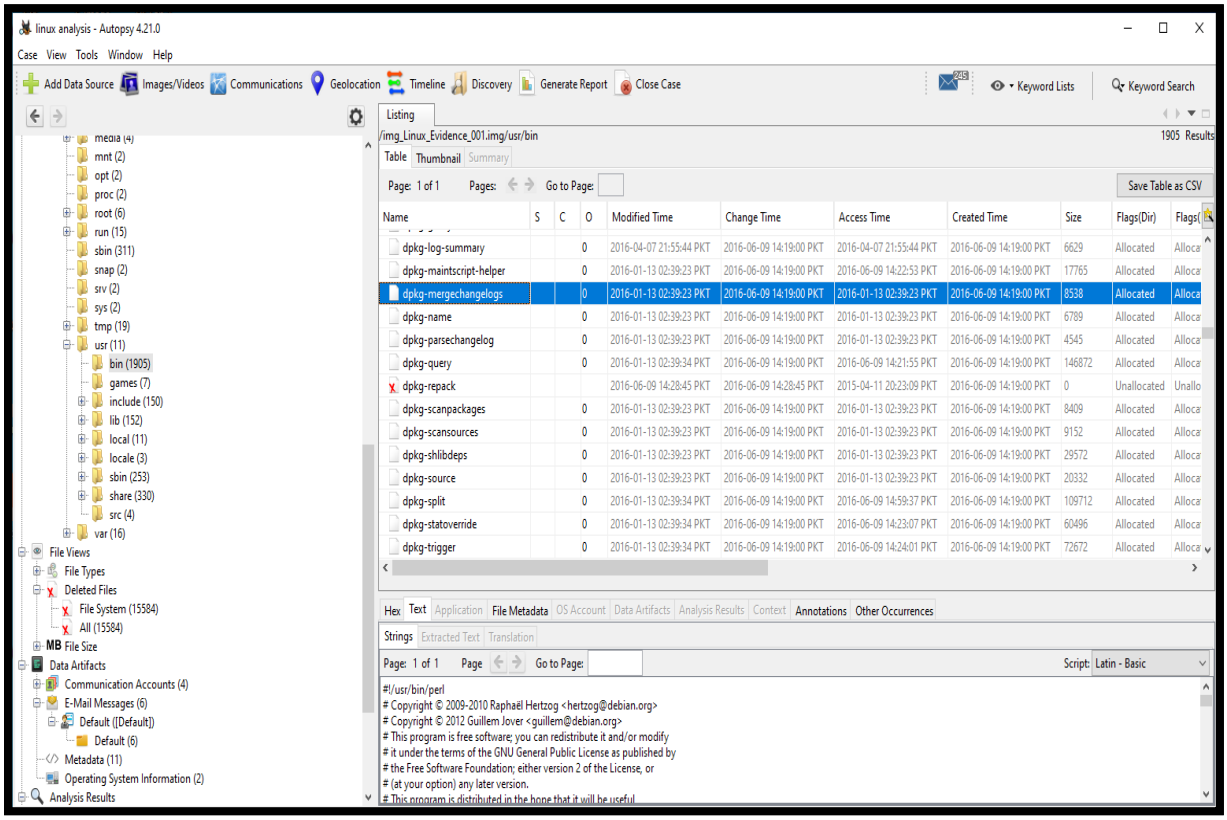
22.Examining Another Image File

Another image file was accessed and analyzed through the Autopsy interface.



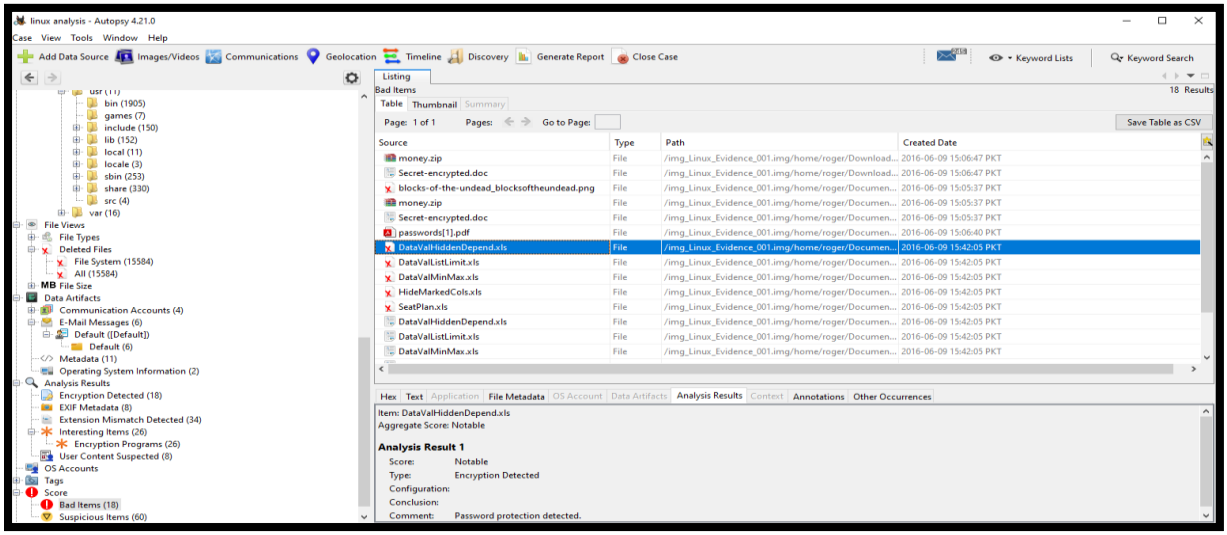
23.Examining an Additional File

An additional file was selected from the forensic image for further examination.



24.Examining a Deleted File

A deleted file was accessed from the forensic image. The available analysis results were examined to identify information associated with the deleted evidence.



Additional Linux Case

Practical Demonstration

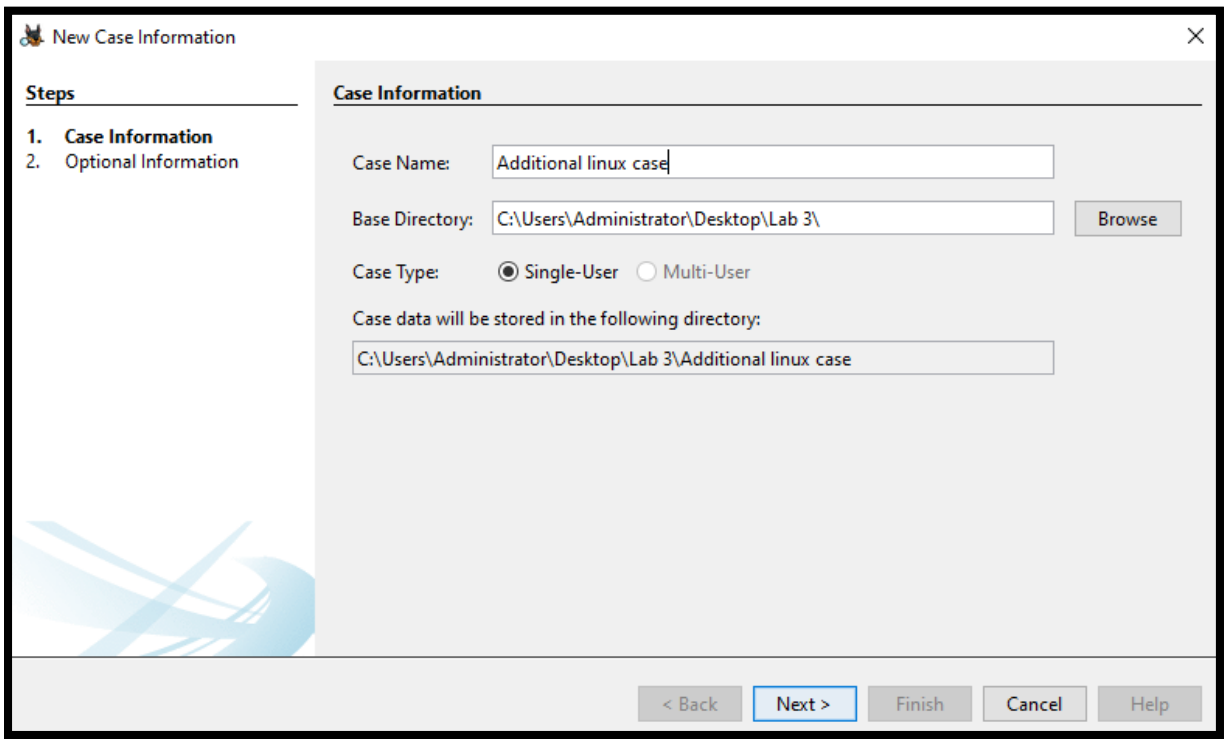
01.Creating a New Case

A new forensic case was created in Autopsy to analyze the additional Linux evidence.



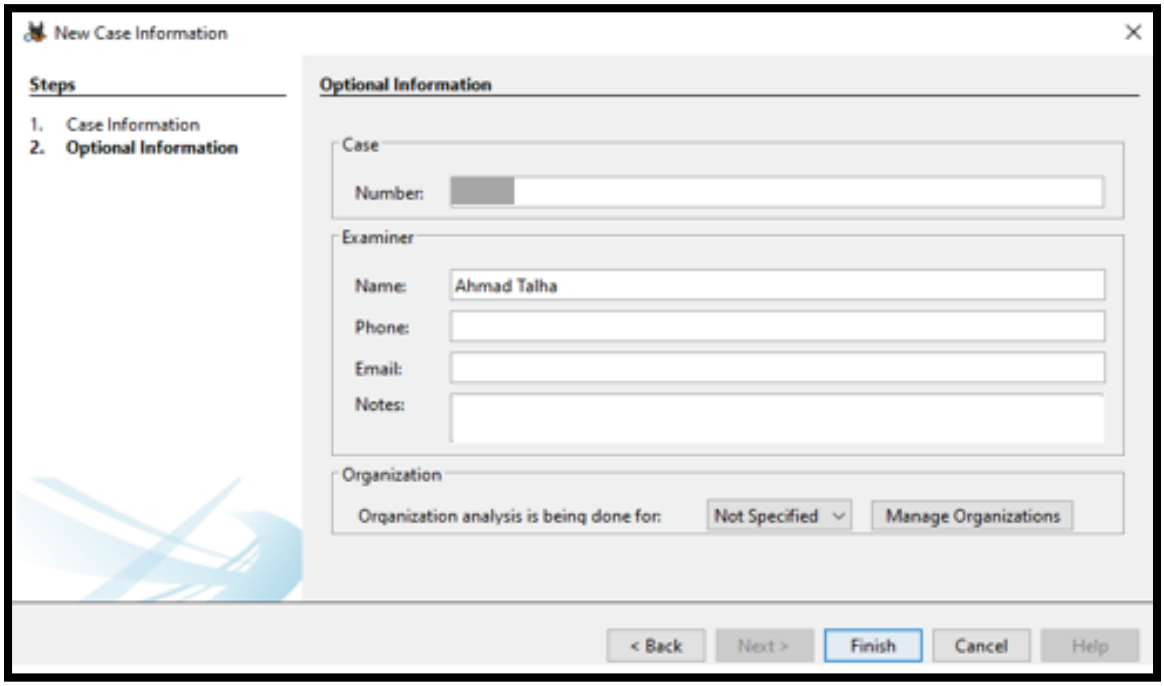
02.Entering Case Information

The required case information was entered to configure the new forensic investigation.



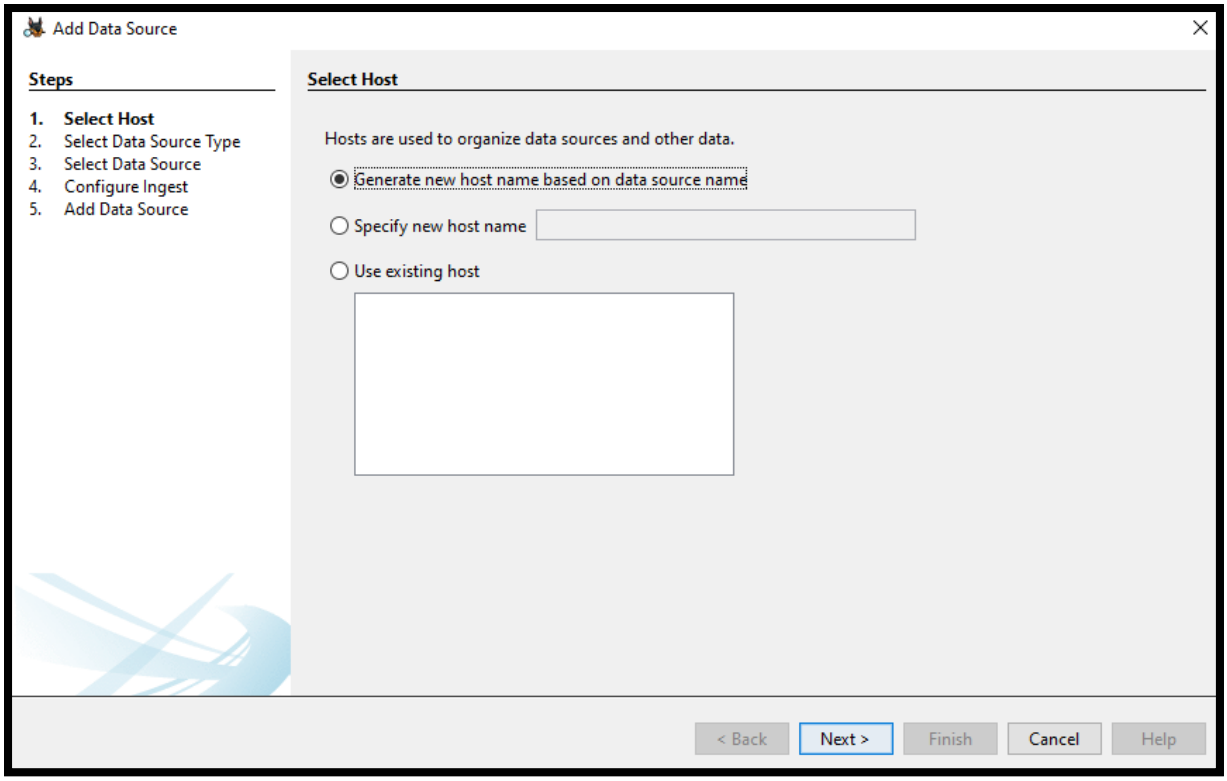
03.Entering Optional Information

The available optional case information was entered before proceeding with the investigation.



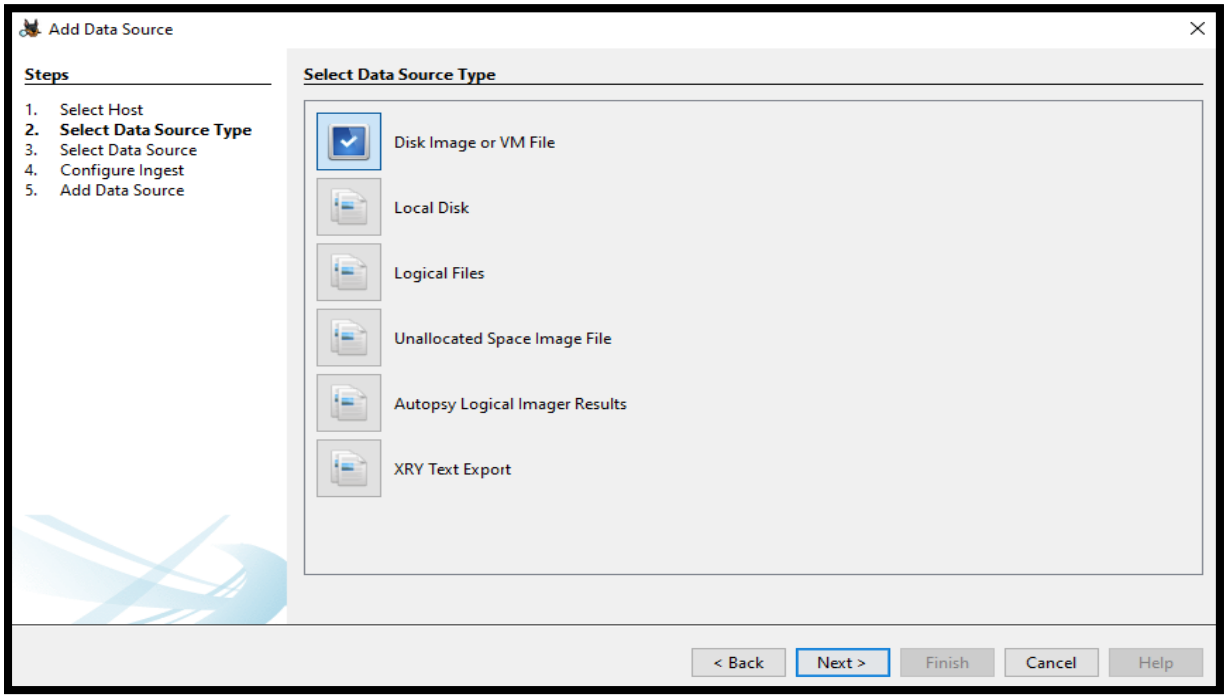
04.Selecting the Host

The appropriate host was selected for the new forensic case.



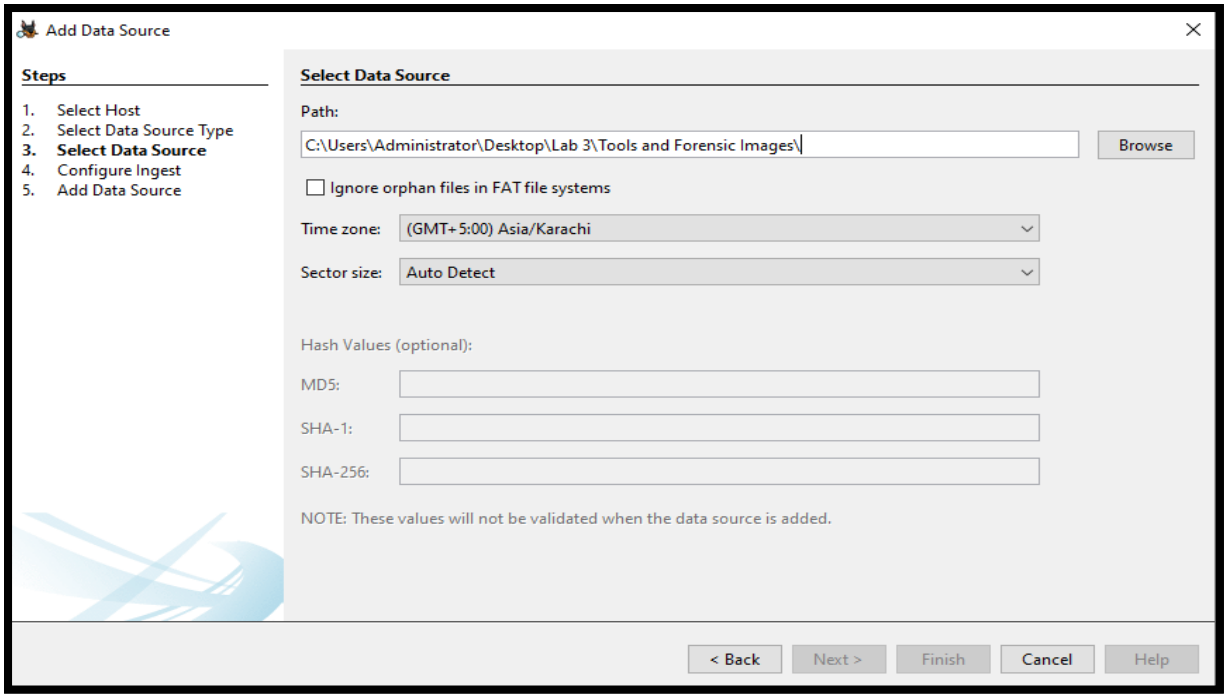
05.Selecting the Data Source Type

The **Disk Image or VM File** option was selected as the data source type.



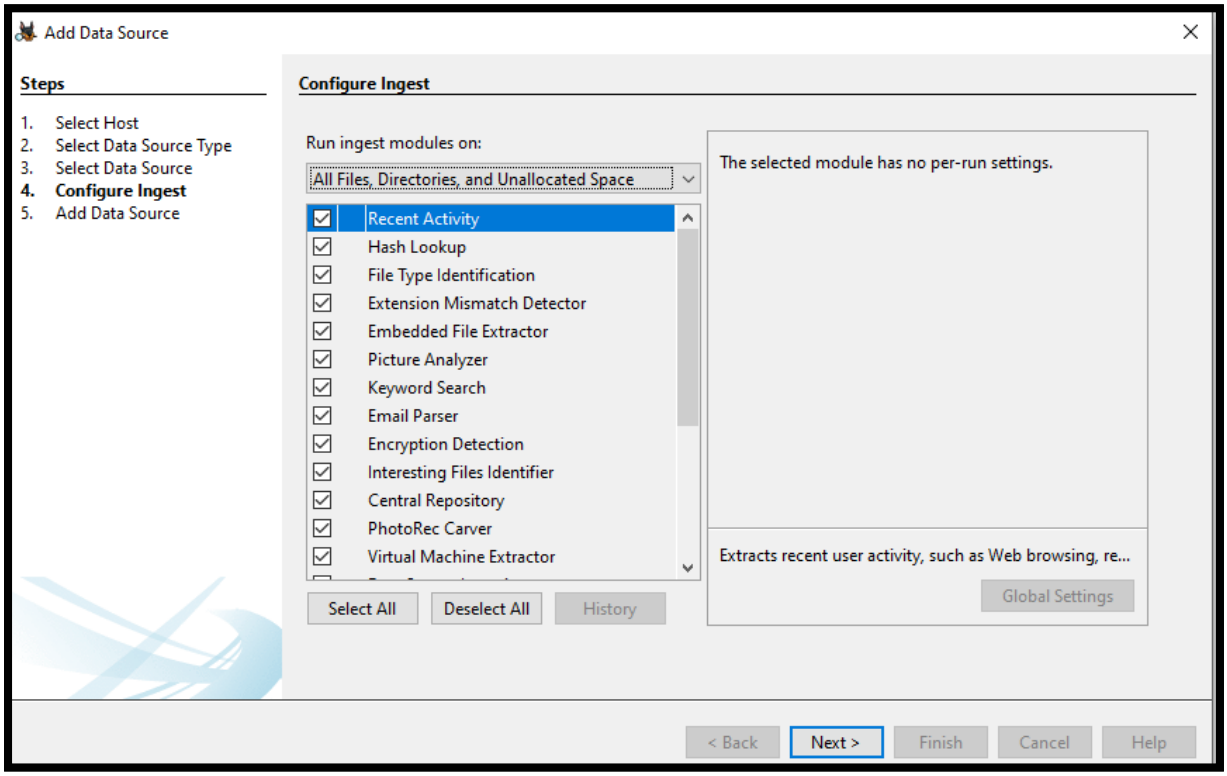
06.Selecting the Data Source

The location of the forensic image was browsed and selected as the data source.



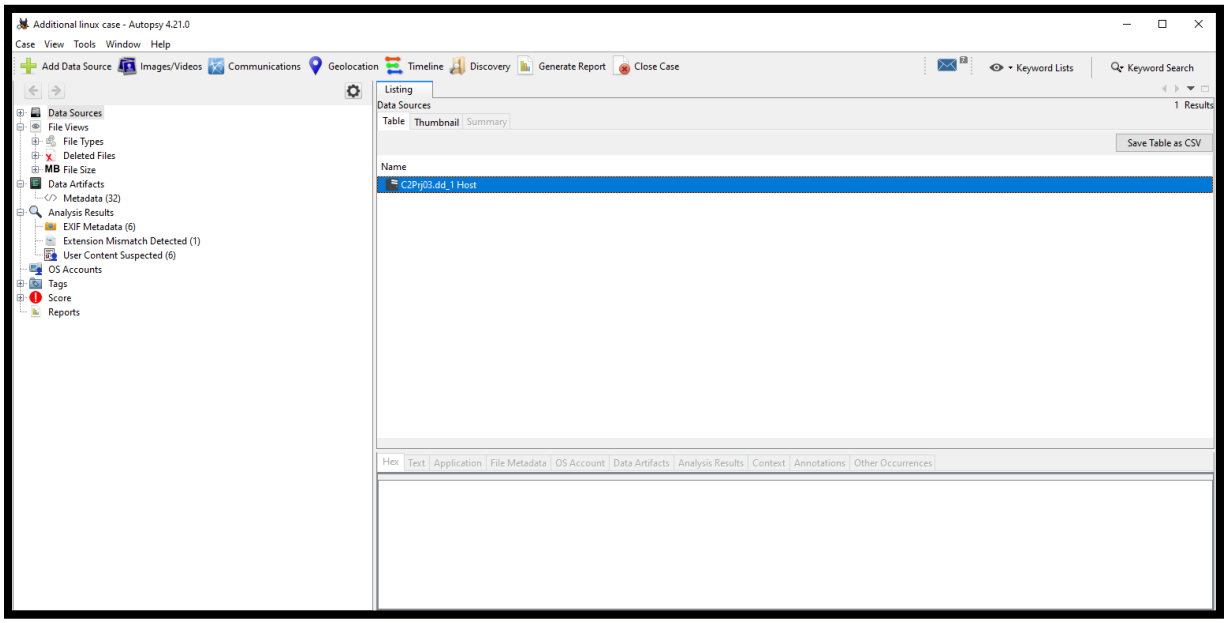
07.Configuring Ingest Modules

The default ingest configuration was retained and the process was continued.



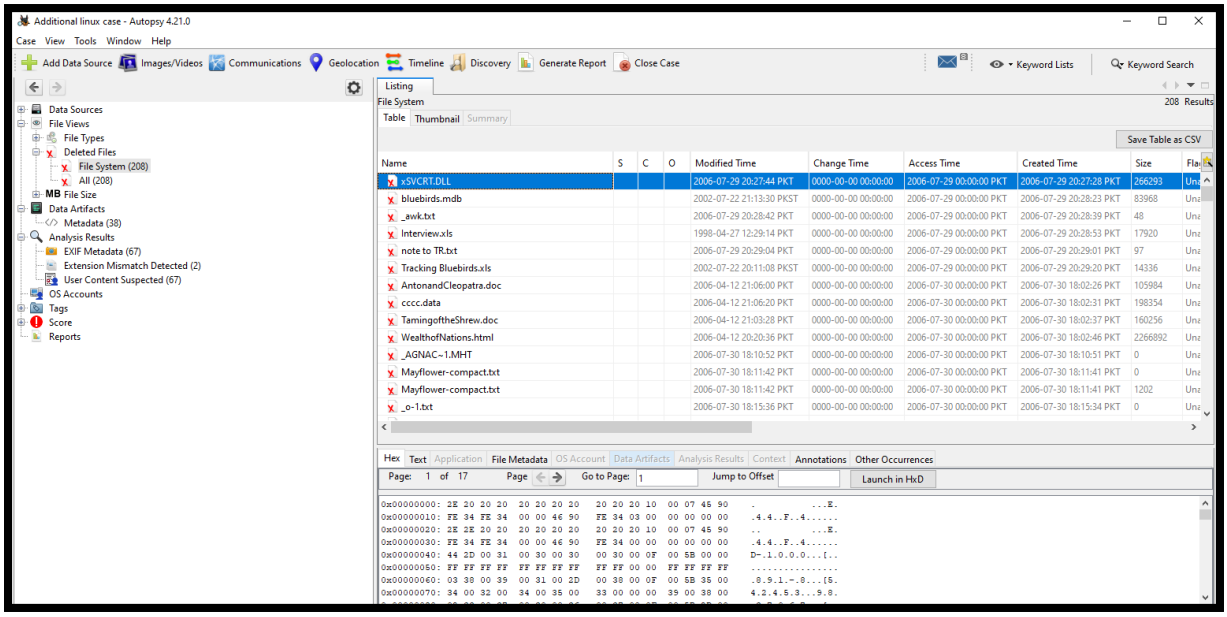
08.Analyzing the Linux File System

After adding the image, Autopsy processed the contents of the additional Linux case and made the file system available for examination.



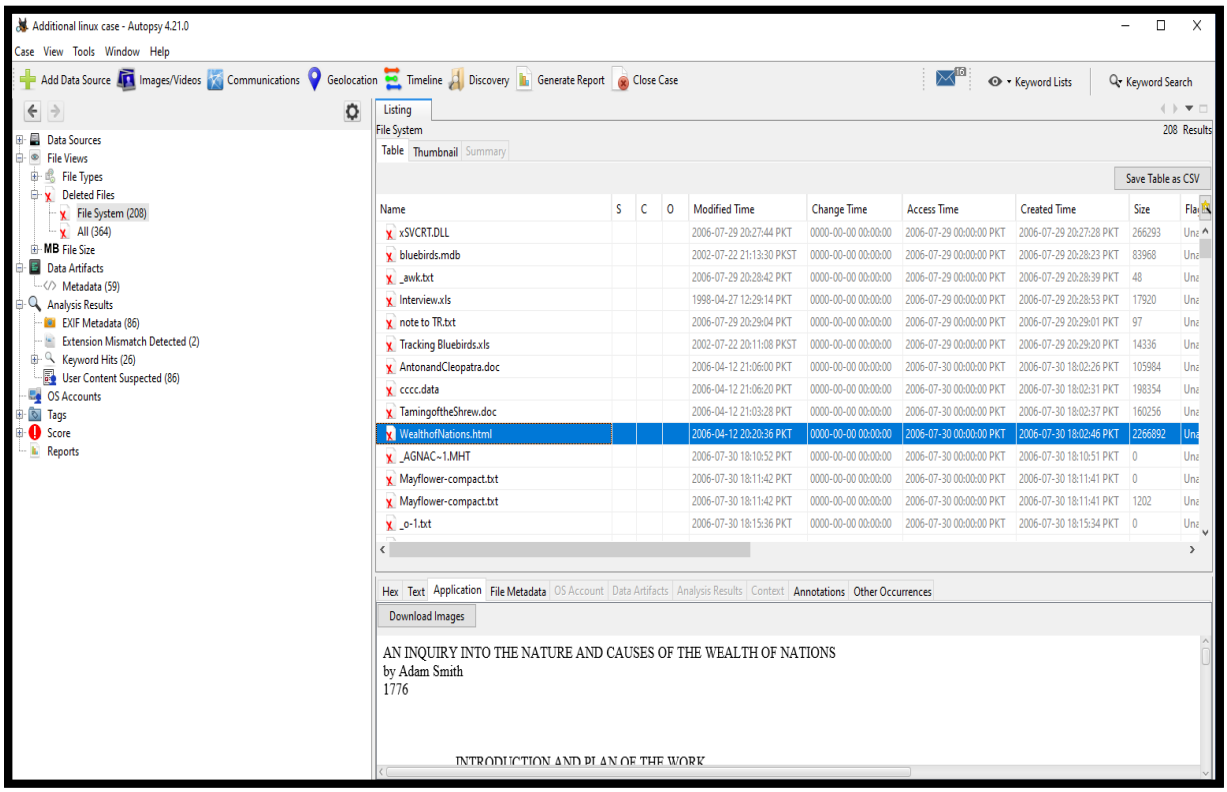
09.Examining the File System

The file system was accessed to examine the available files, including deleted files identified by Autopsy.



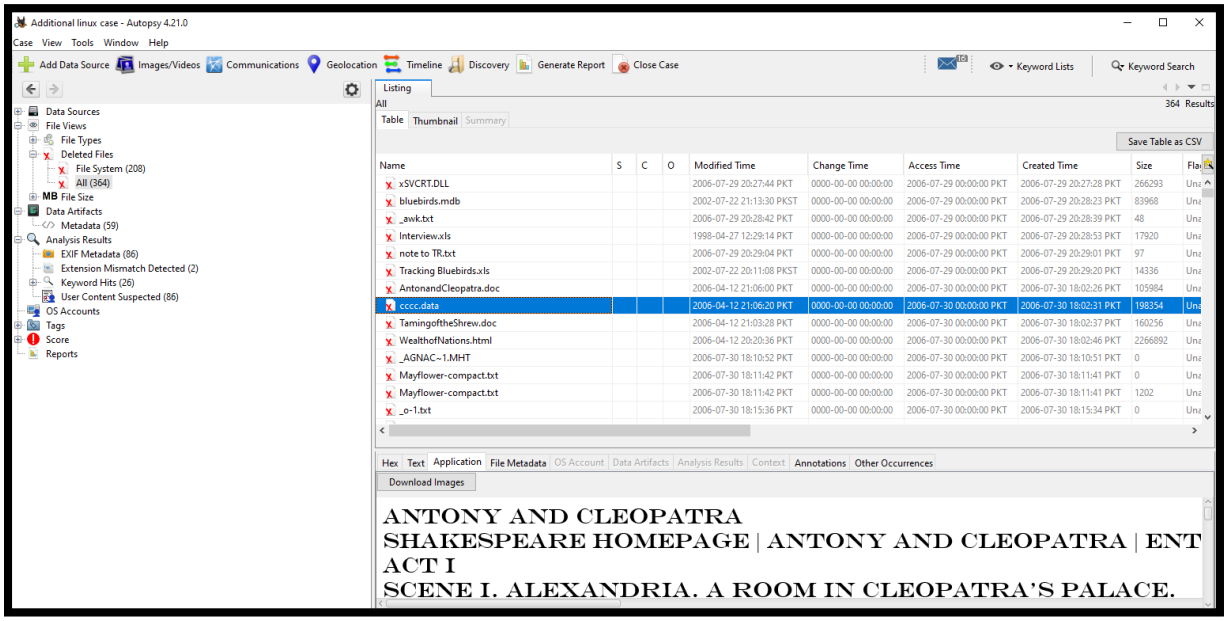
10.Examining an HTML File

An HTML file was selected and its available content was examined through Autopsy.



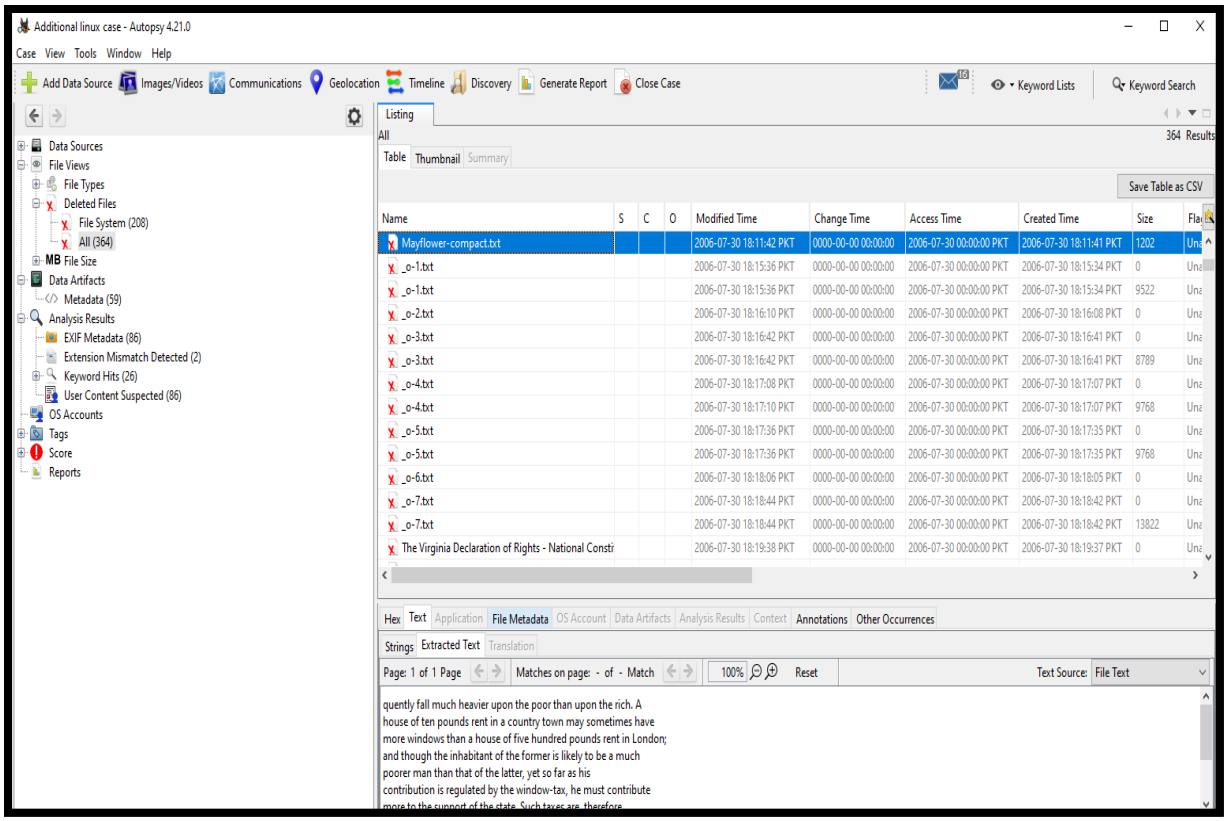
11.Examining a DATA File

A DATA file was selected to examine the information and content associated with it.



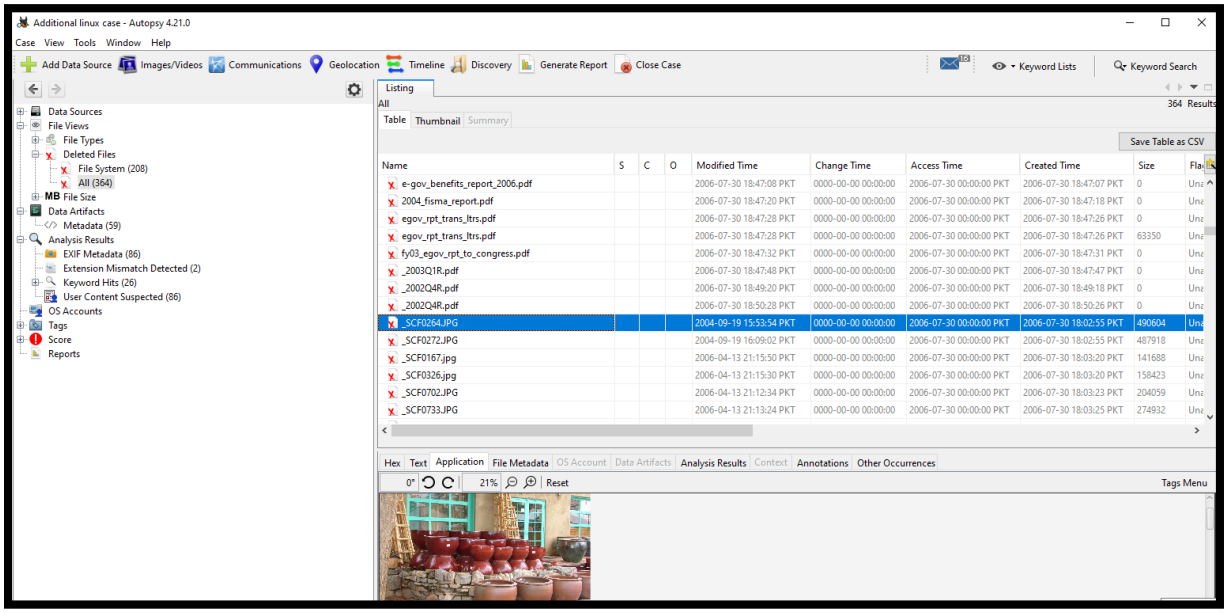
12.Examining a TXT File

A TXT file was selected and its metadata was reviewed through the Autopsy interface.



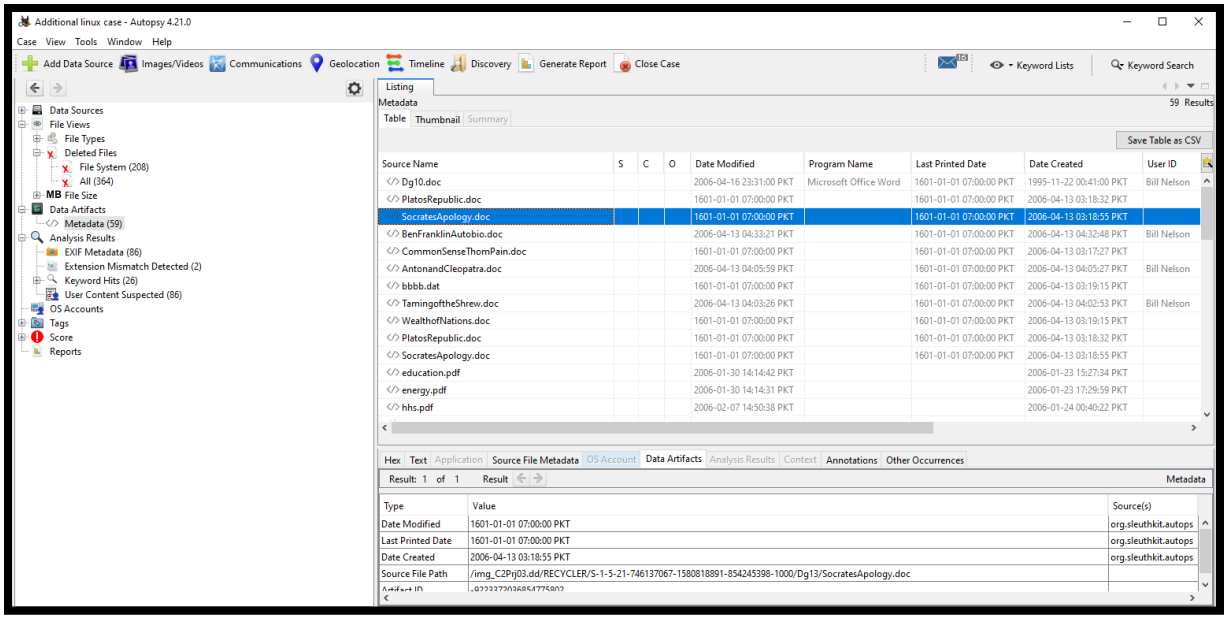
13.Examining a JPG File

A JPG image was selected and opened to examine its contents.



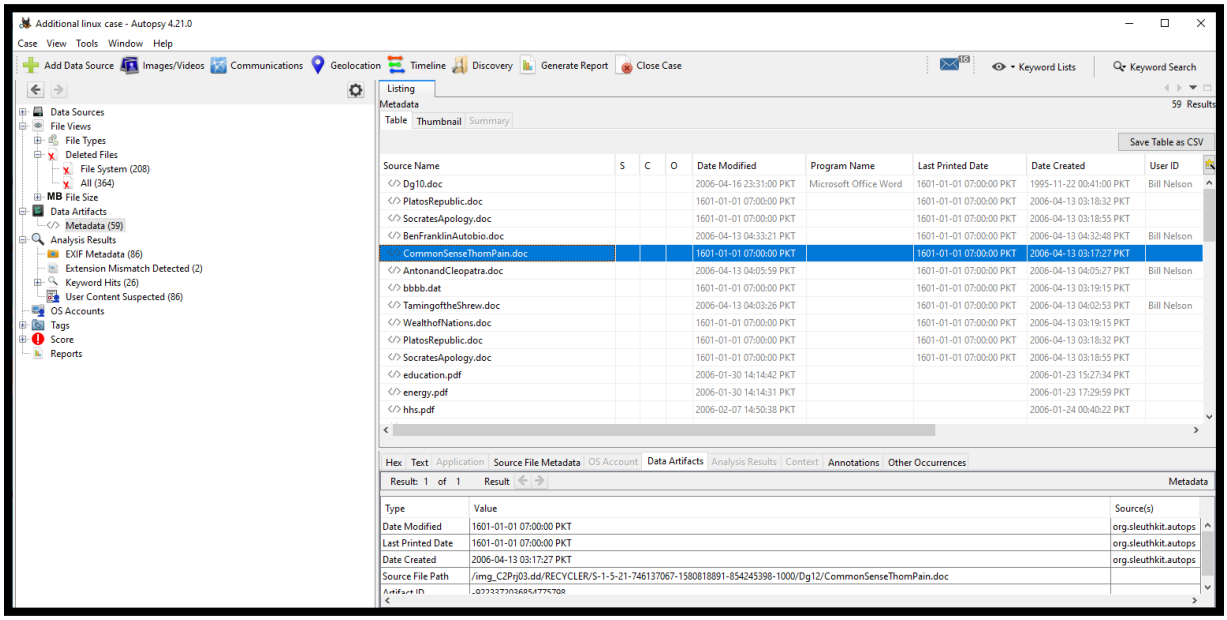
14.Examining a DOC File

A DOC file was selected and its available data artifacts were examined.



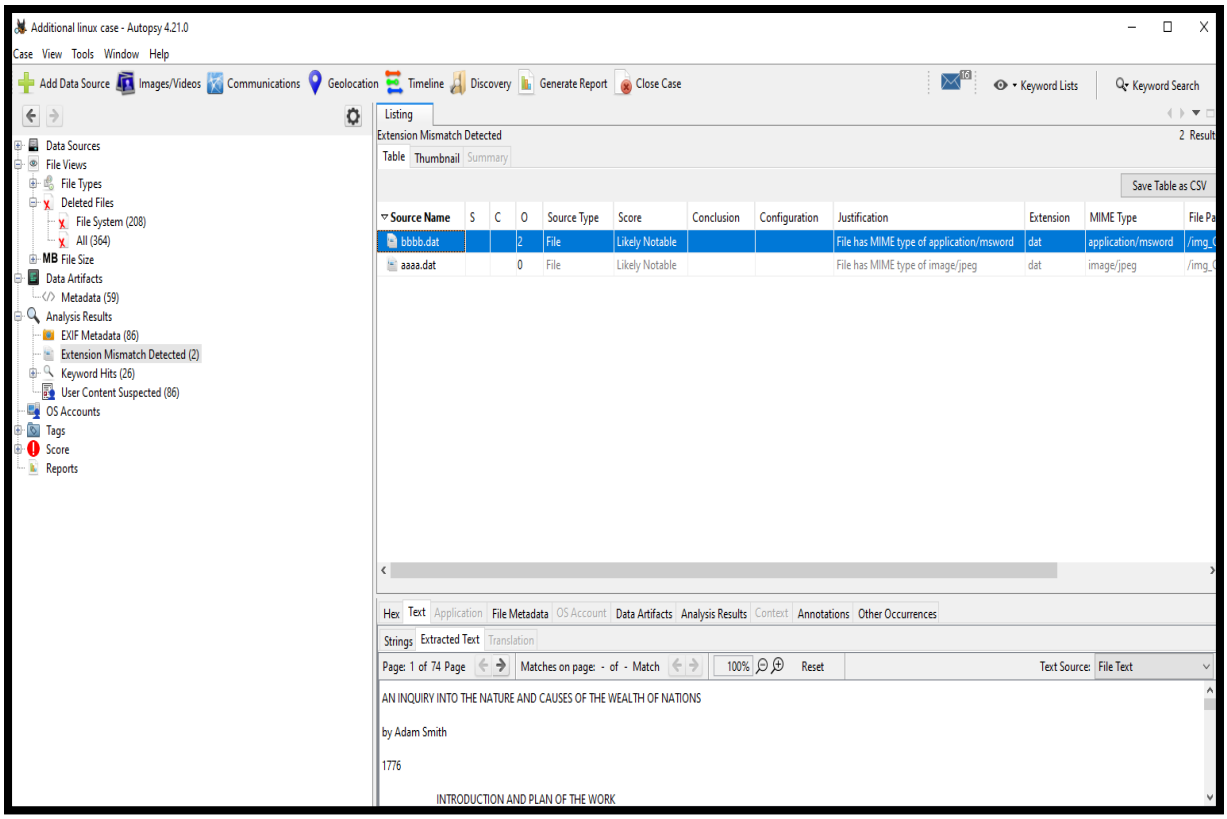
15.Examining Another DOC File

Another DOC file was selected to examine its available data artifacts.



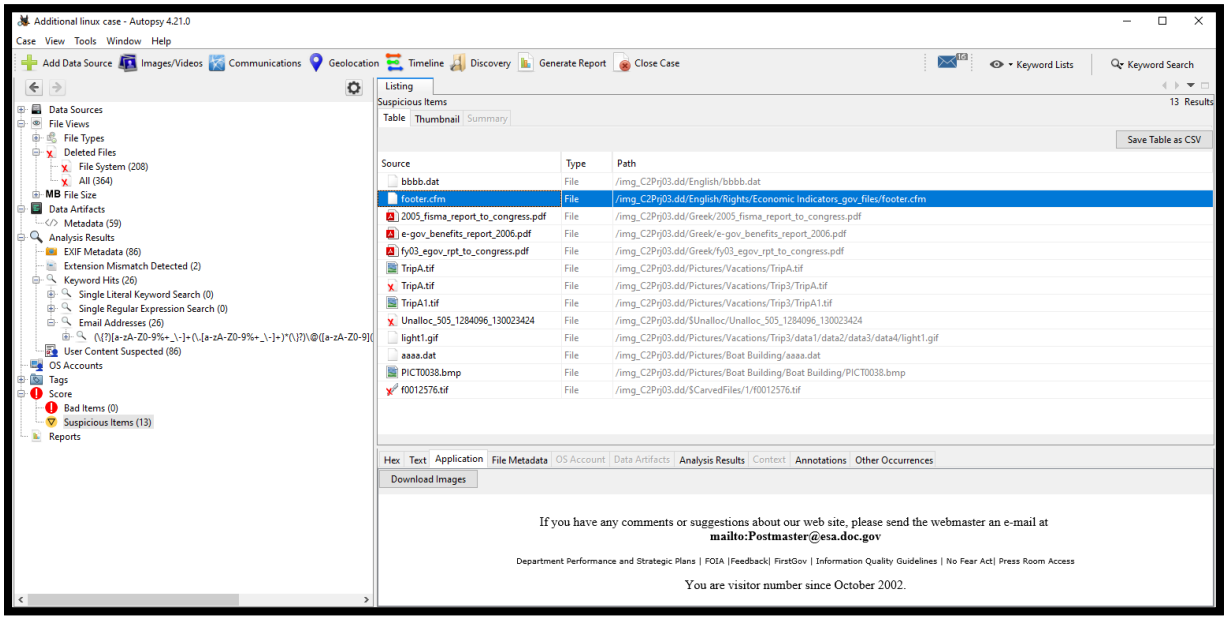
16.Examining a DAT File

A DAT file was selected from the analysis results and its extracted text was examined.



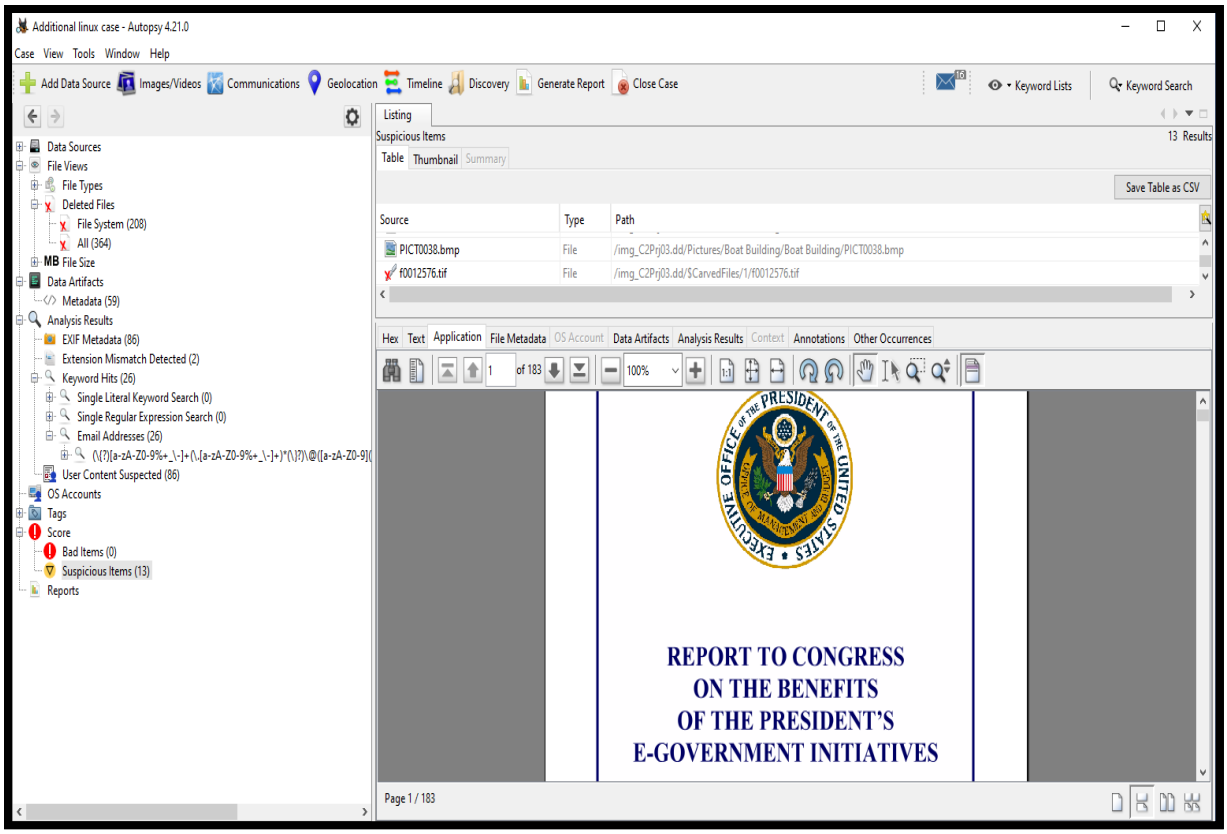
17.Examining a CFM File

A CFM file was selected and its available content was examined through Autopsy.



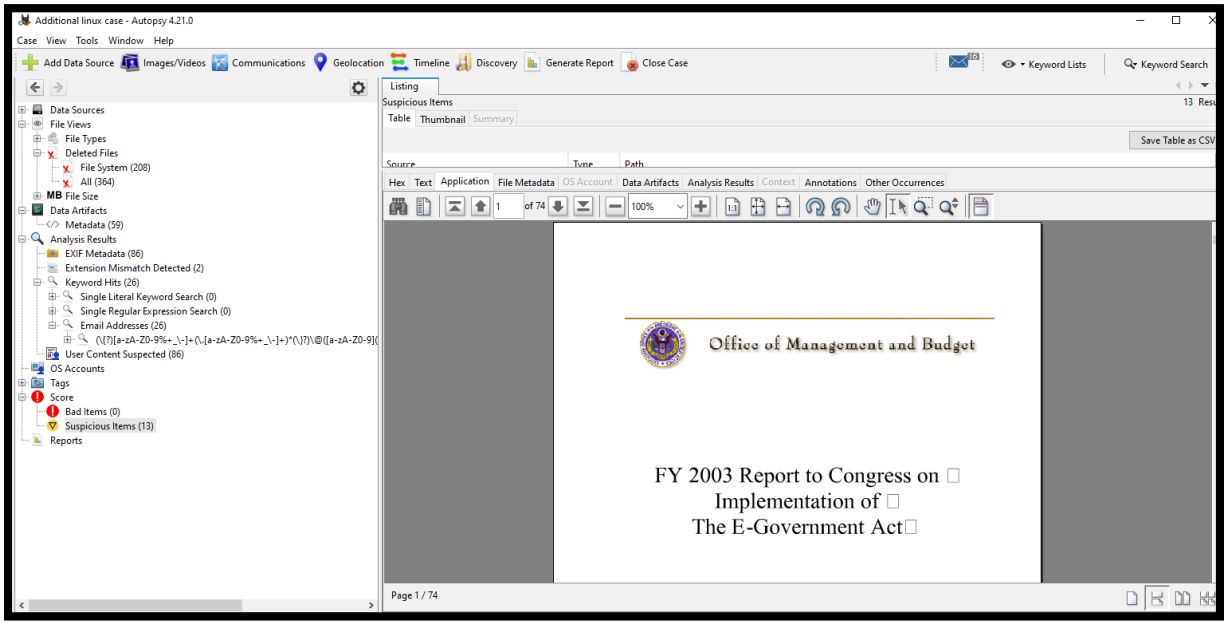
18.Examining Suspicious Items

The **Suspicious Items** section was accessed and a file was selected for further examination.



19.Examining Another Suspicious File

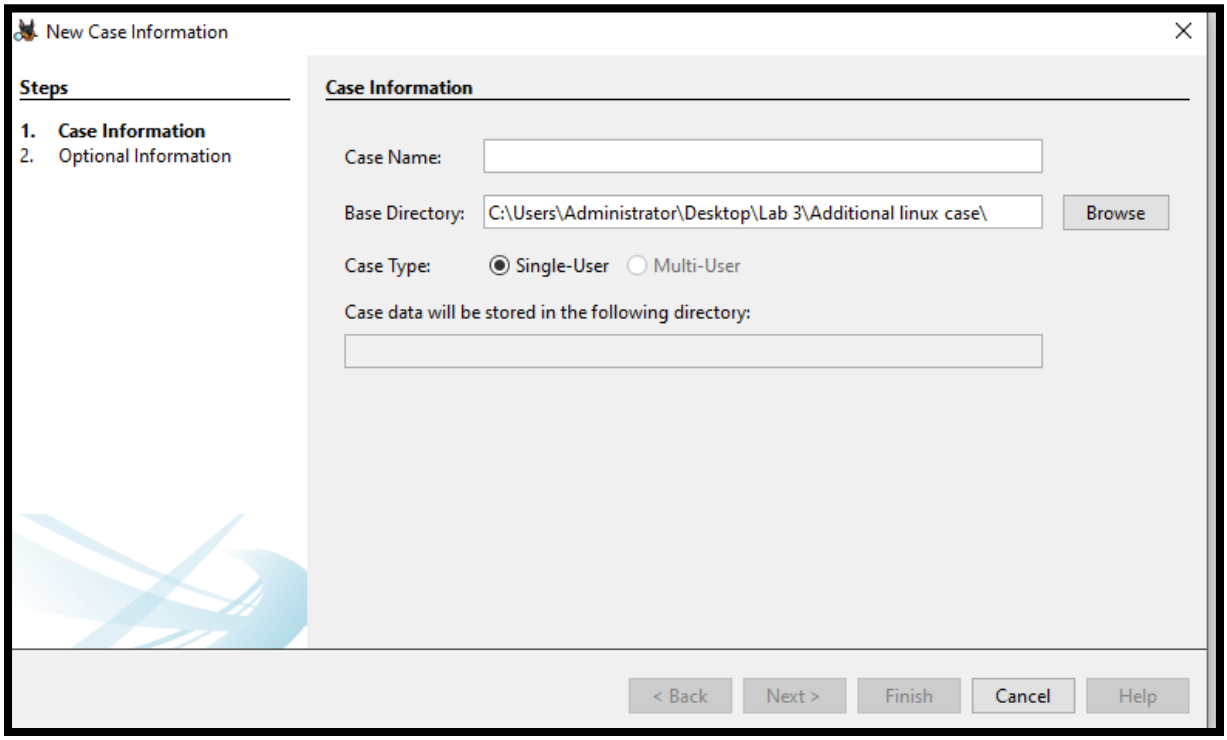
Another file from the **Suspicious Items** section was selected and analyzed.



Additional Forensic Image Analysis

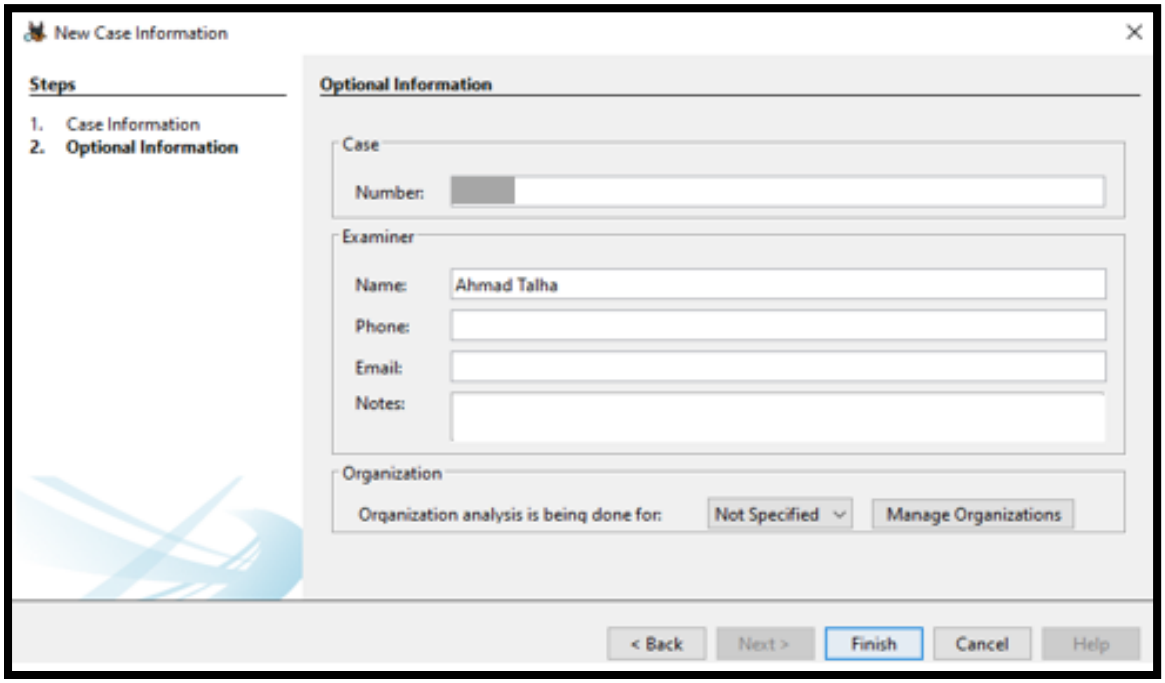
20.Creating a New Case

A new forensic case was created using another forensic image for additional analysis.



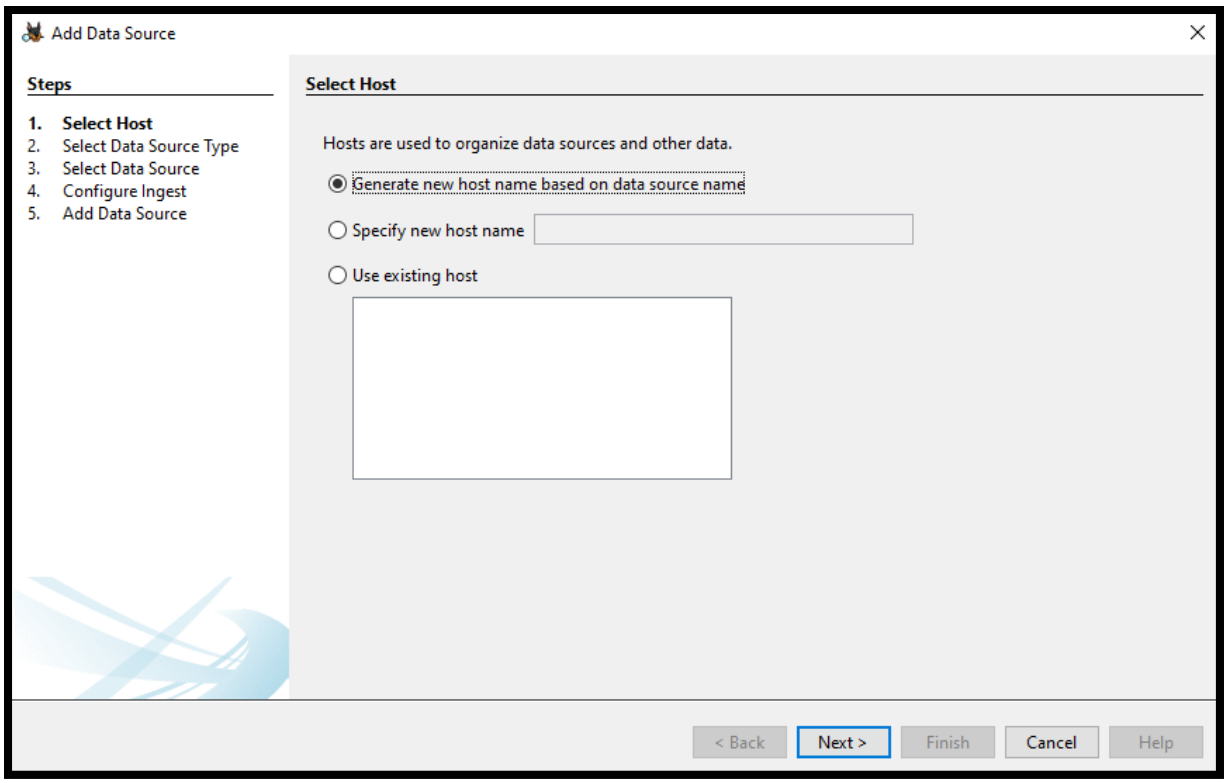
21.Entering Optional Information

The available optional information was entered for the new forensic case.



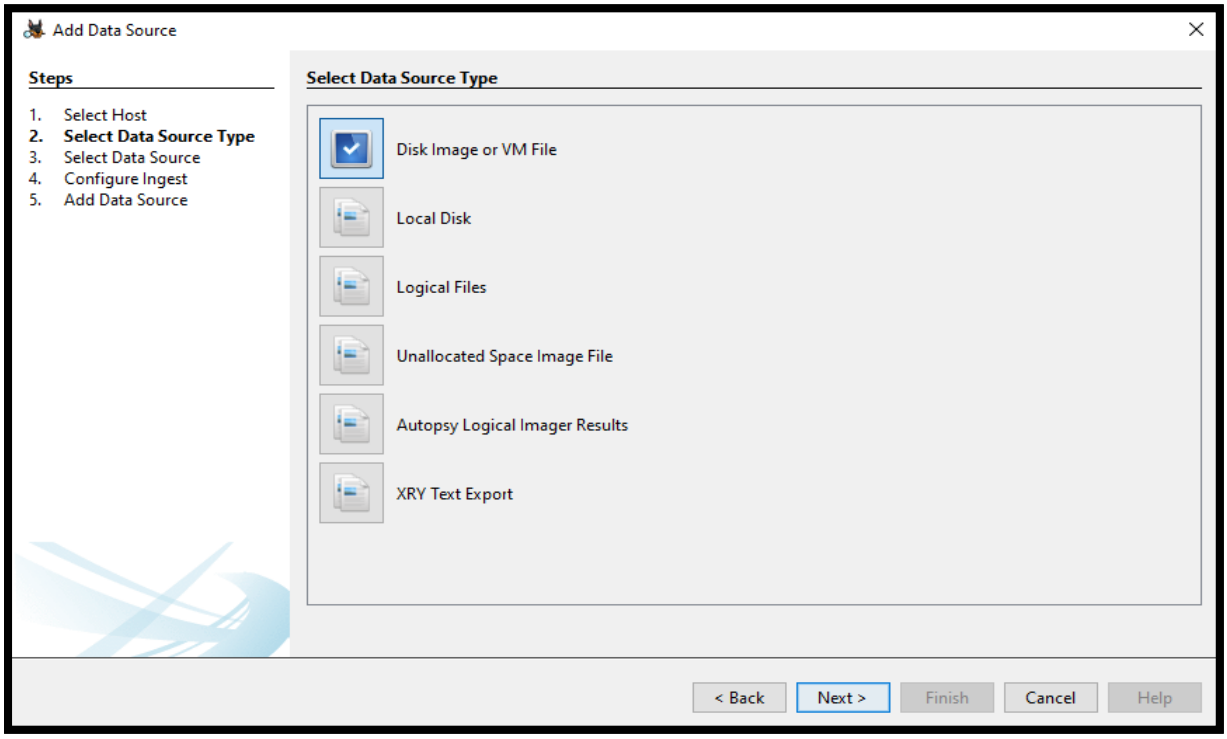
22.Selecting the Host

The appropriate host was selected for the forensic investigation.



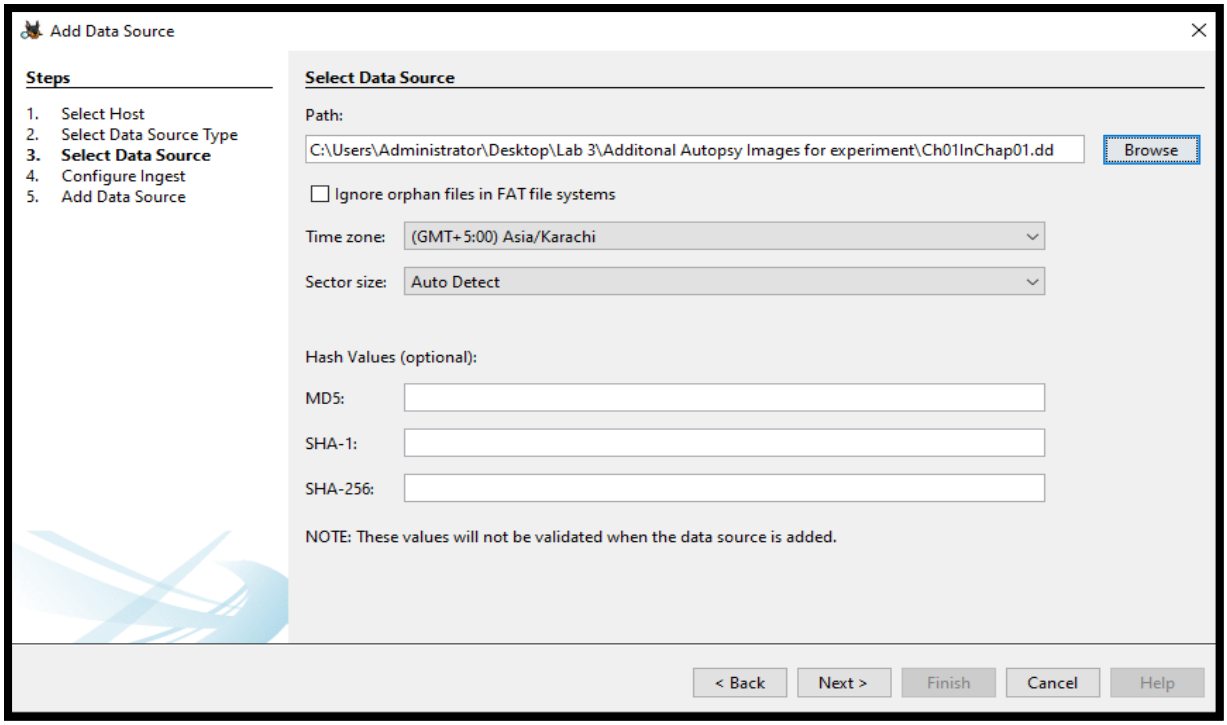
23.Selecting the Data Source Type

The **Disk Image or VM File** option was selected as the data source type.



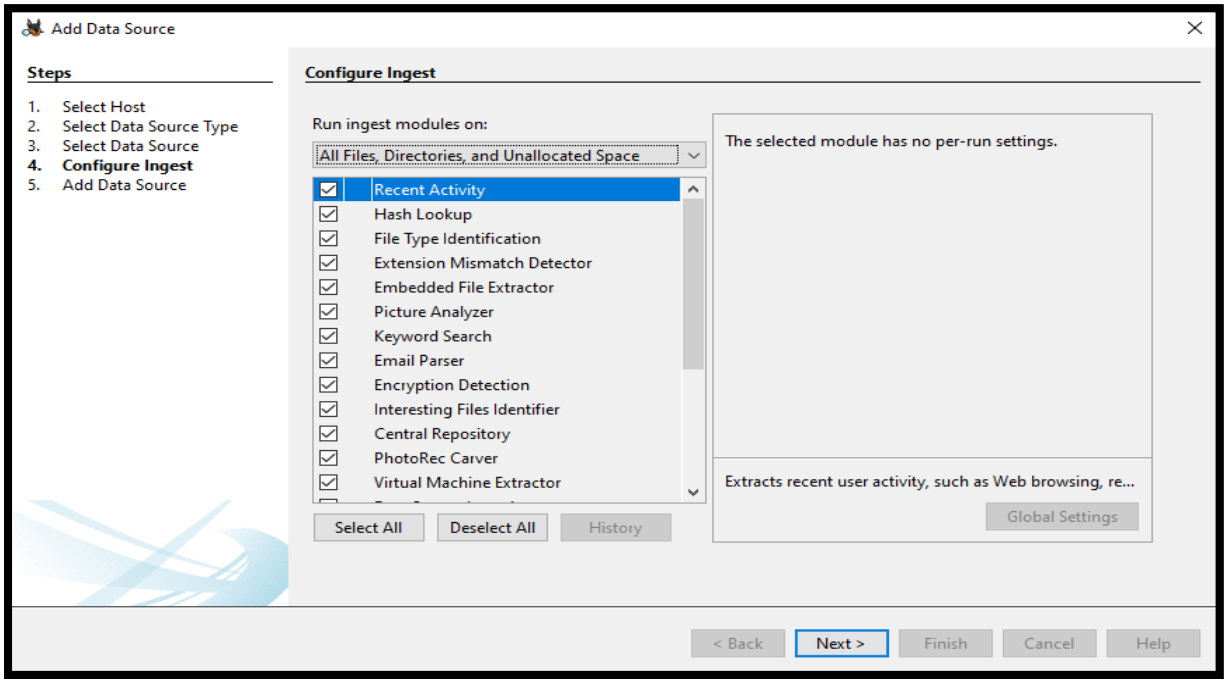
24.Selecting the Data Source

The forensic image was browsed and selected as the data source.



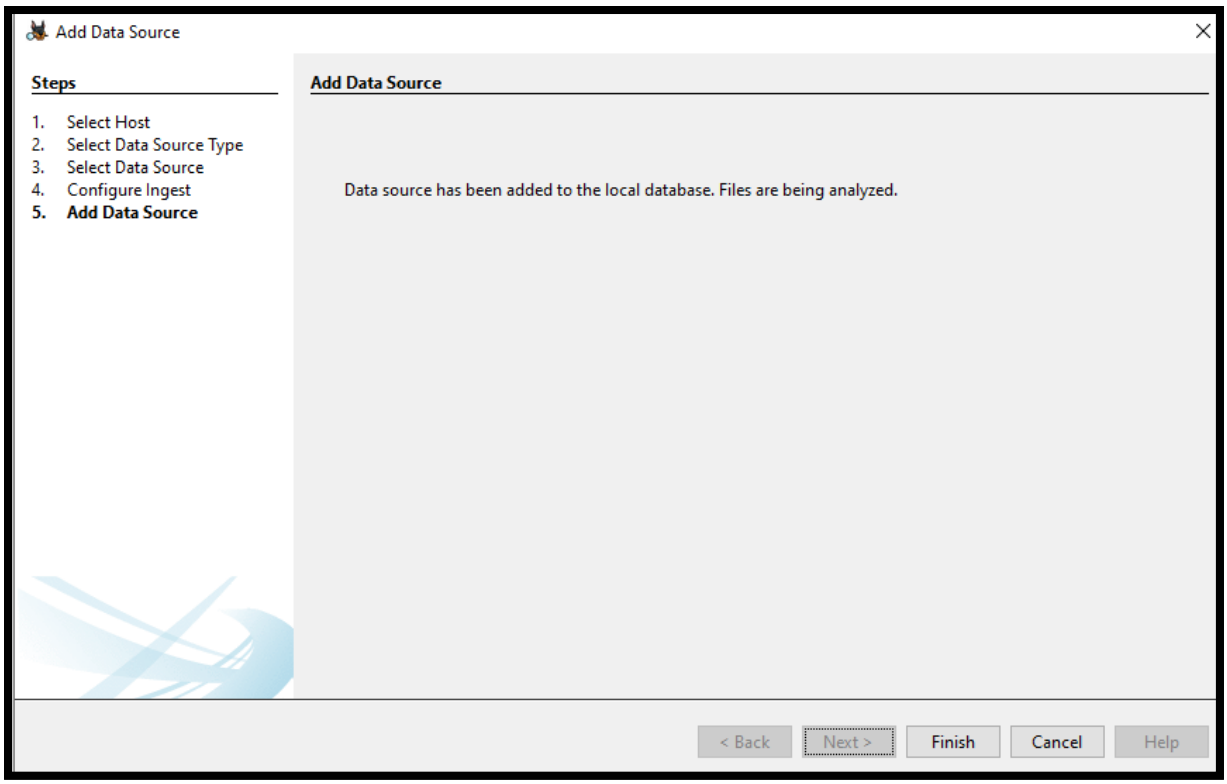
25.Configuring Ingest Modules

The default ingest configuration was retained and the process was continued.



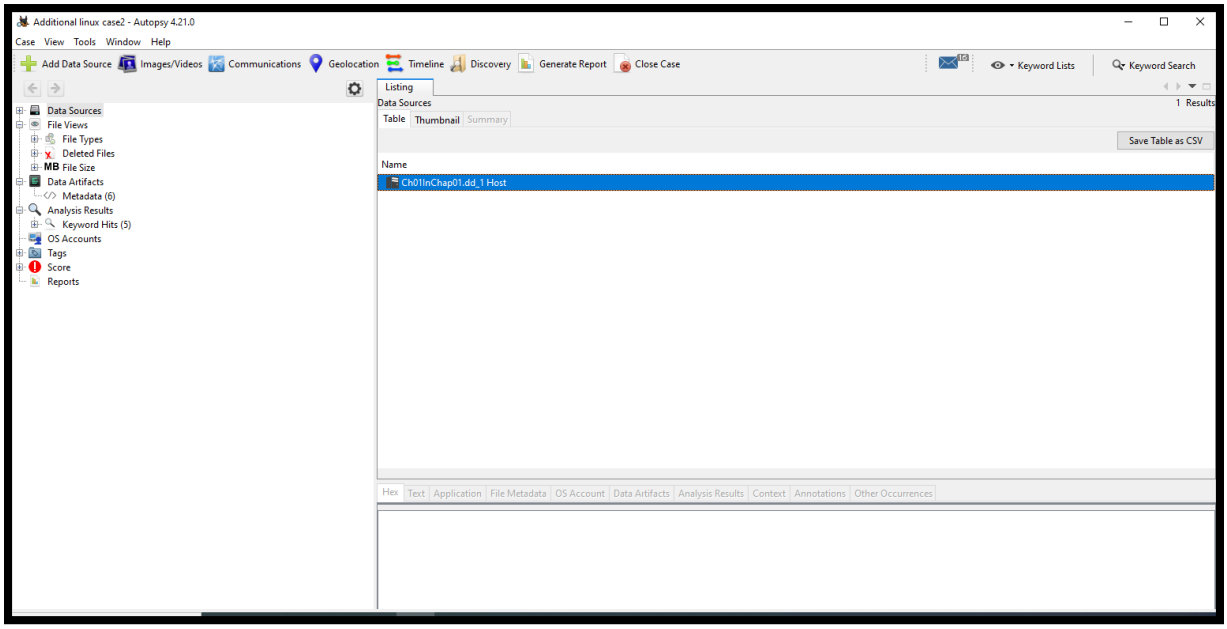
26.Adding the Data Source

The forensic image was successfully added to the new case.



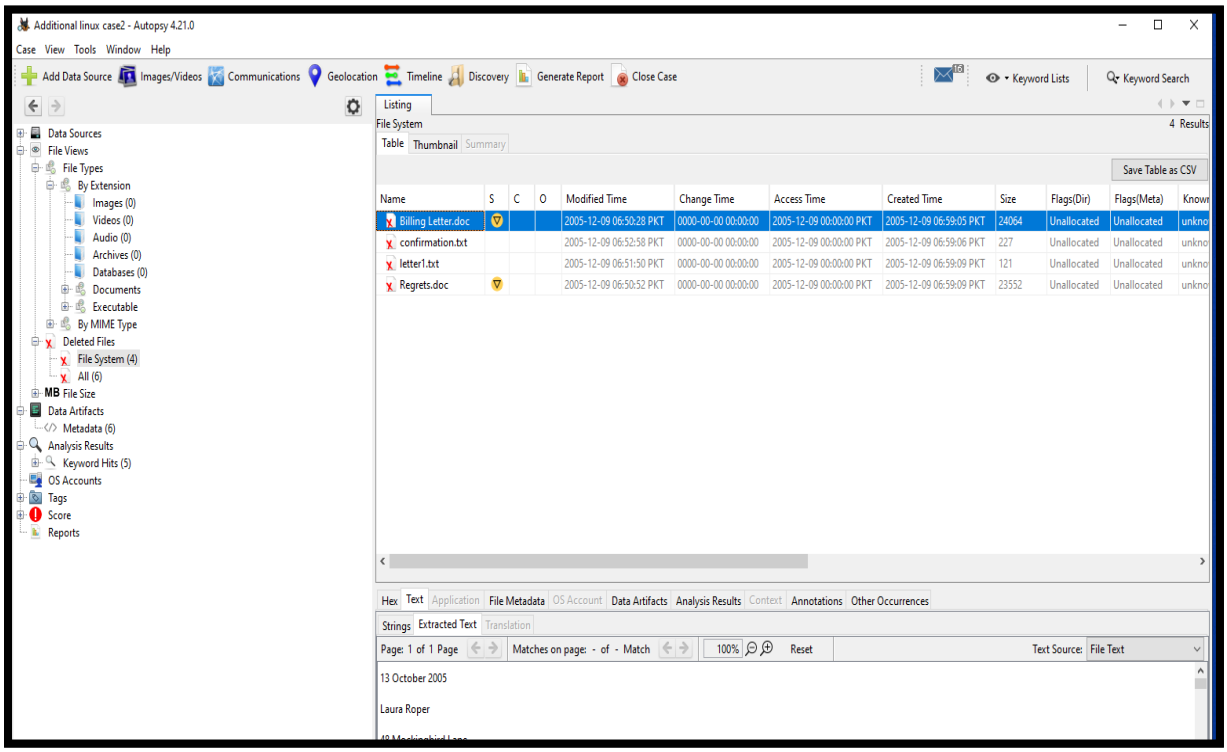
27.Analyzing the Forensic Image

After adding the .dd image, Autopsy processed its contents and made the file system available for examination.



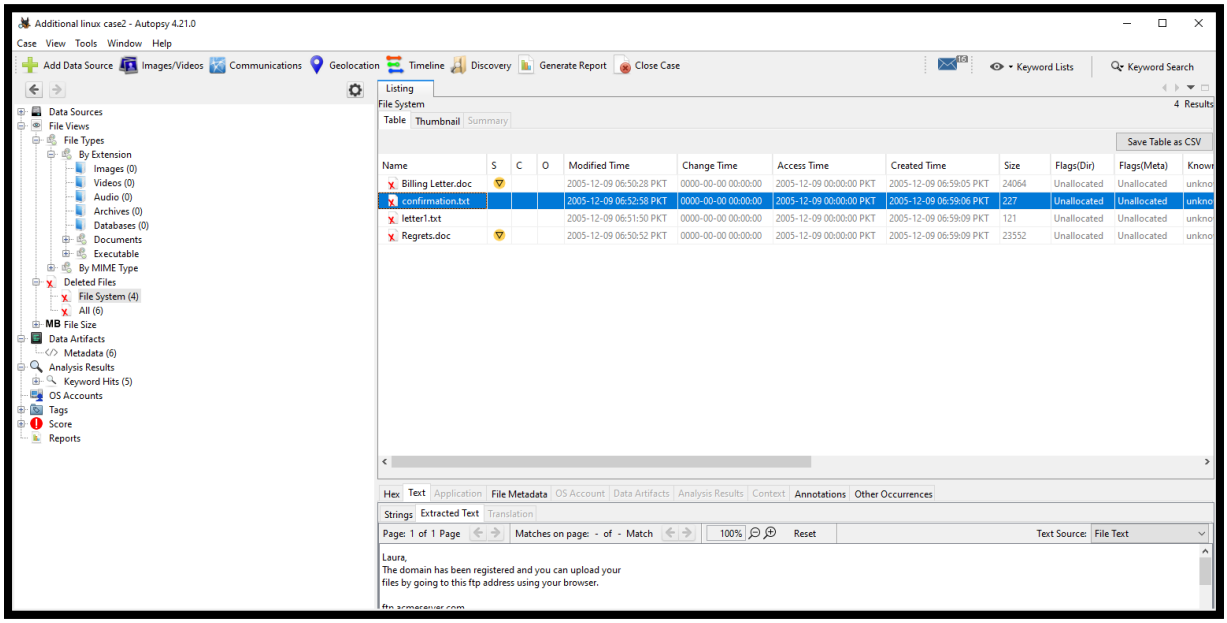
28.Examining Deleted Files

The deleted files section was accessed to identify deleted evidence present within the forensic image.



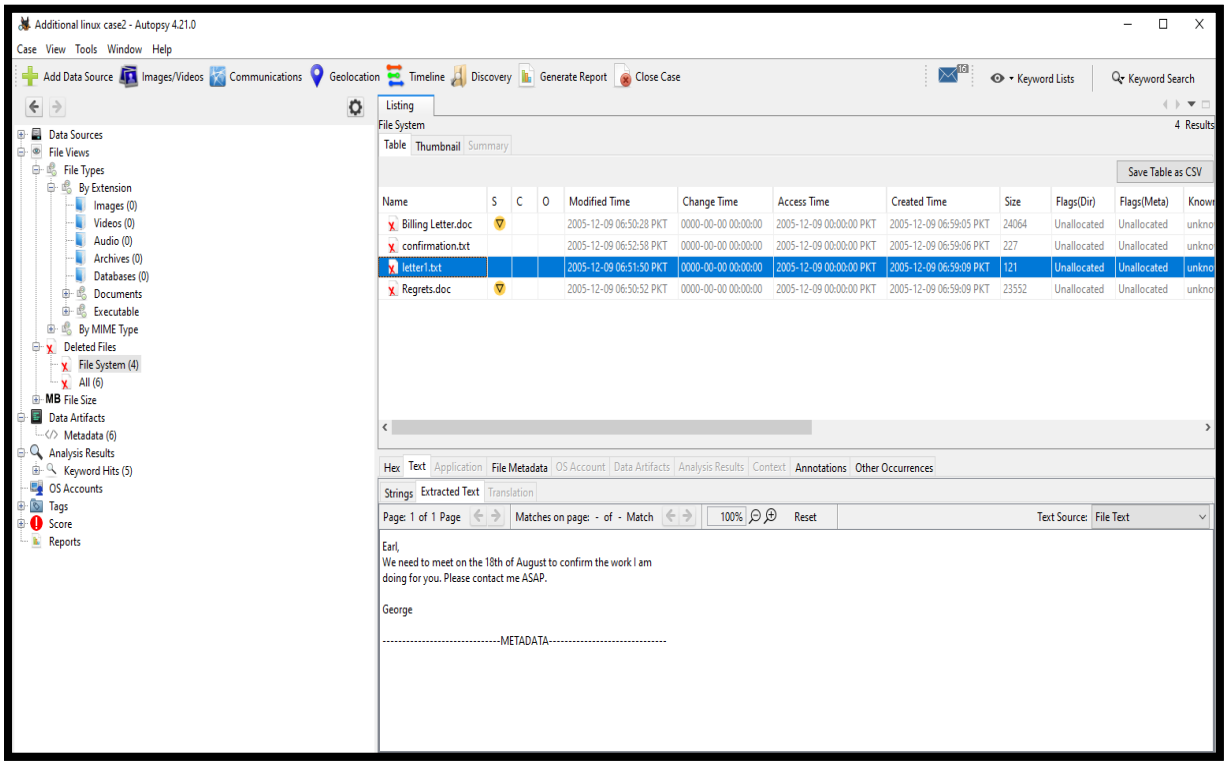
29.Examining a TXT File

A TXT file was selected from the deleted files and its extracted data was examined.



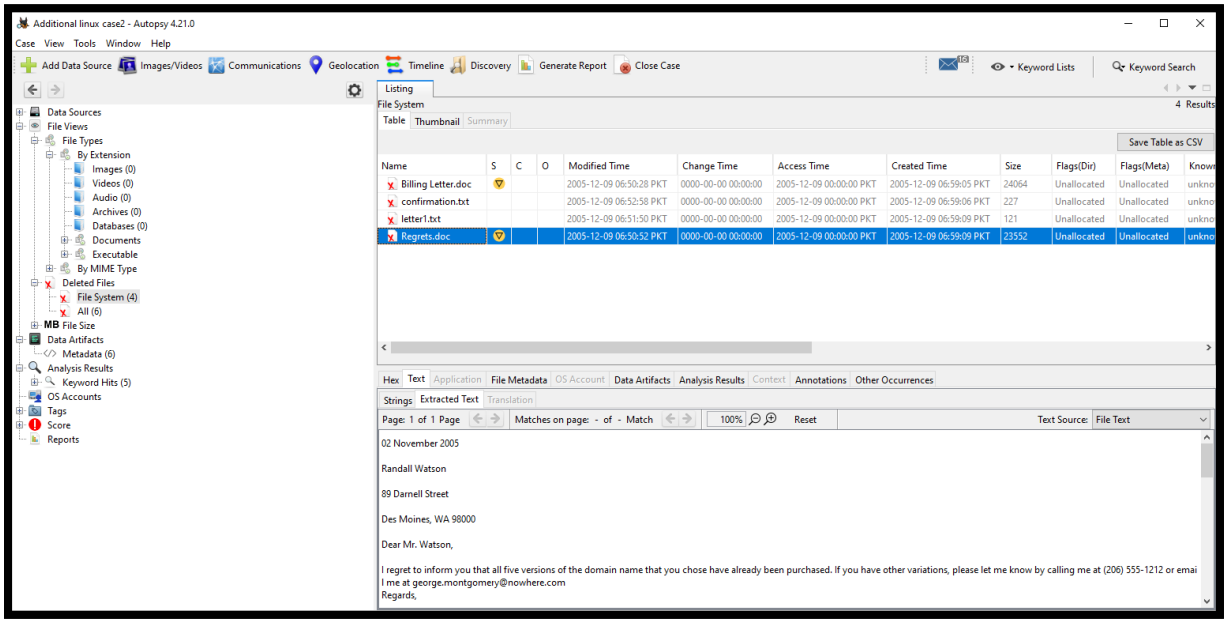
30.Examining Extracted Text

The extracted text of the selected TXT file was reviewed to examine the information recovered from the evidence.



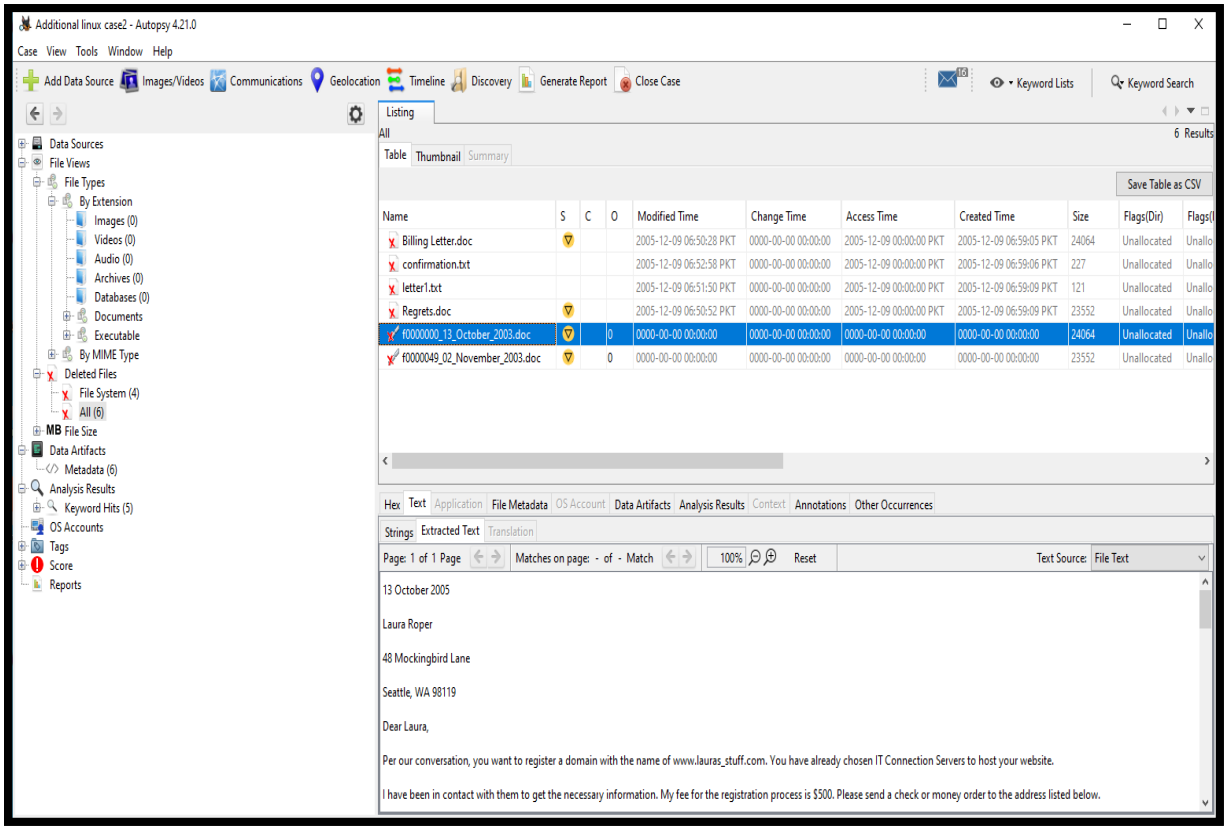
31.Examining Data from a DOC File

The extracted data associated with a DOC file was examined through Autopsy.



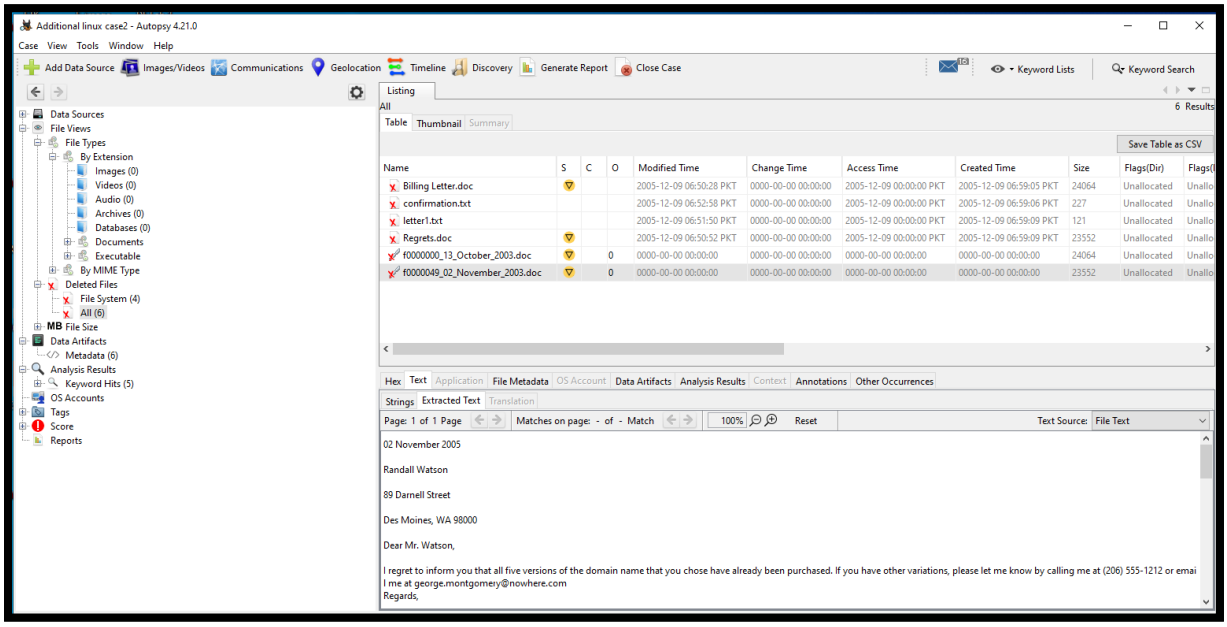
32. Examining Data from Another DOC File

Another DOC file was selected and its extracted data was reviewed.



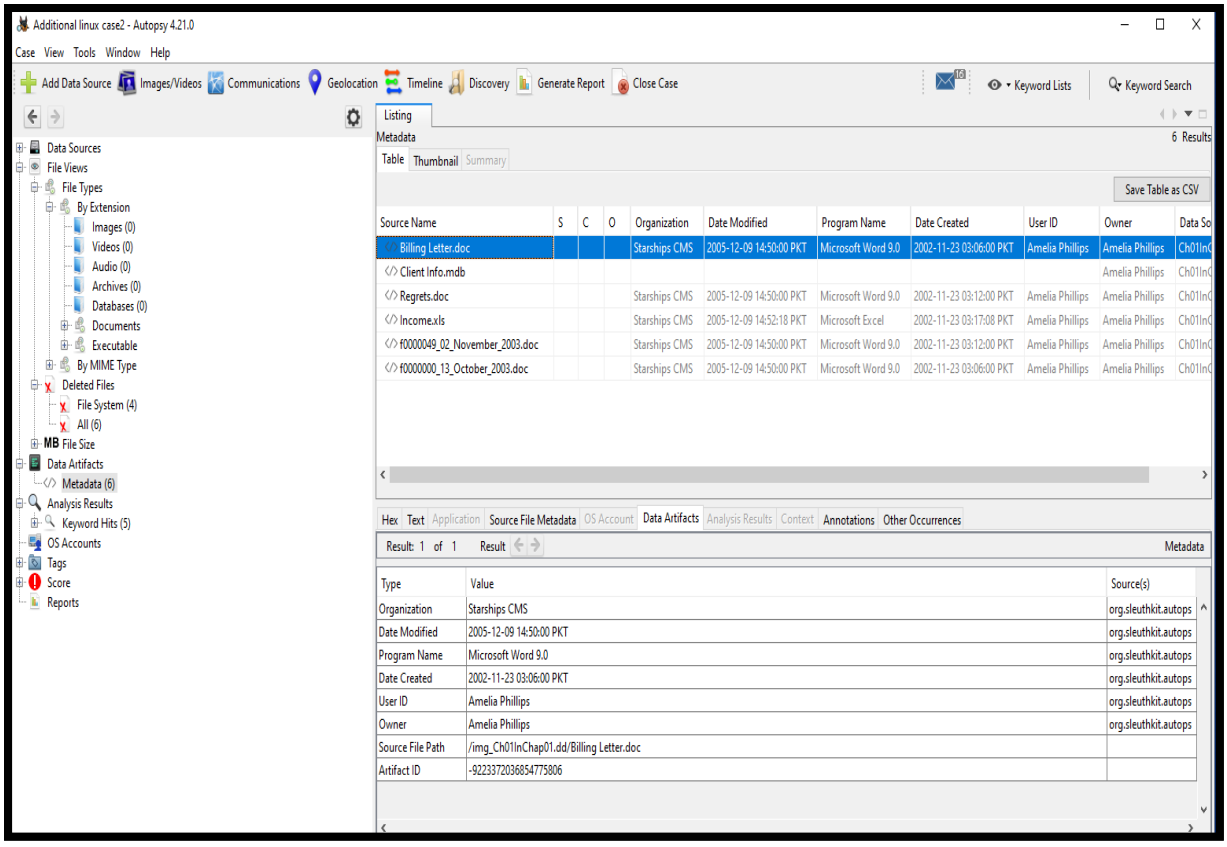
33.Examining Additional DOC File Data

An additional DOC file was examined to review its extracted information.



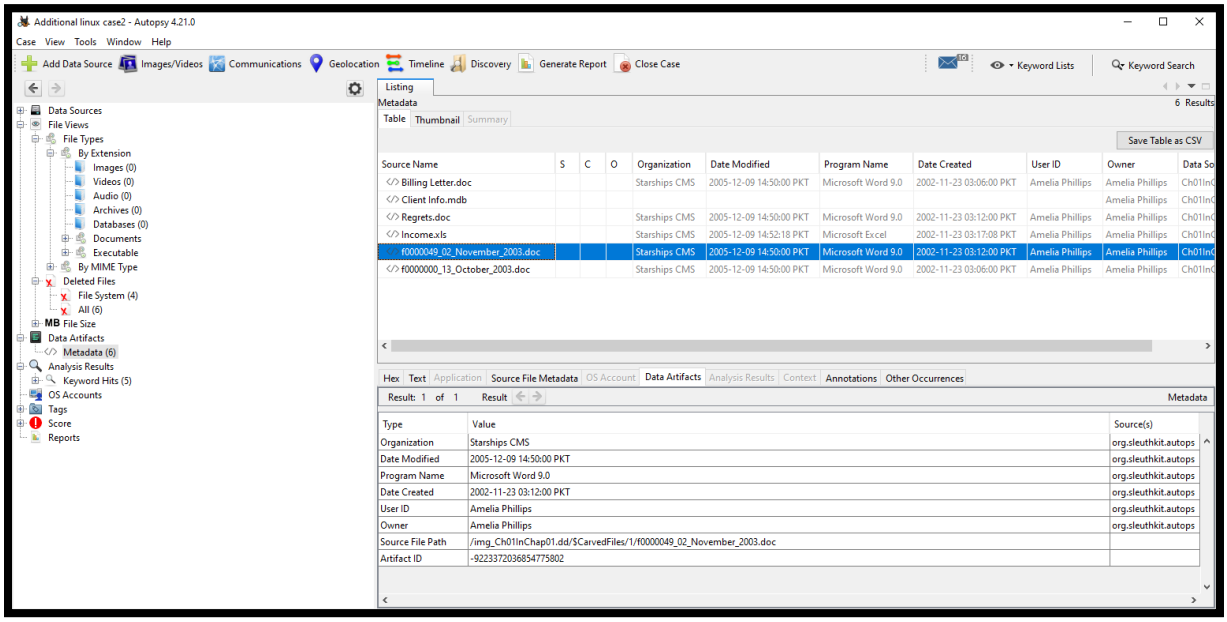
34.Examining DOC File Metadata

The data artifacts and metadata associated with a DOC file were examined.



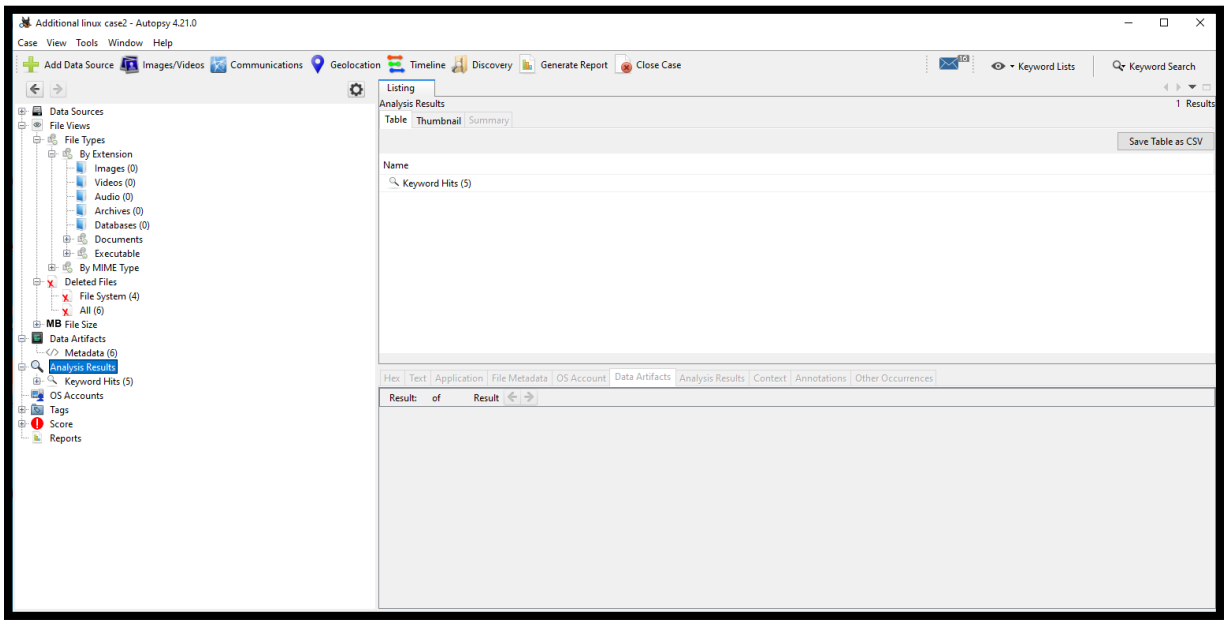
35.Examining Additional DOC Metadata

Additional metadata and data artifacts associated with a DOC file were reviewed.



36.Examining Keyword Hits

The **Keyword Hits** section was examined to identify files or evidence containing the configured search terms.



The Sleuth Kit (TSK)

Introduction

The Sleuth Kit (TSK) is a collection of command-line digital forensic tools used for detailed analysis of forensic disk images and file systems. It provides tools for examining partitions, file systems, metadata structures, files, directories, deleted files, and other low-level forensic information.

Purpose of The Sleuth Kit

The main purpose of The Sleuth Kit is to provide detailed file-system and disk-image analysis through command-line utilities.

In this lab, TSK was used to:

- Analyze the partition structure of the forensic image
- Identify the file system type
- Examine forensic image information
- Analyze NTFS metadata
- Examine Master File Table (MFT) entries
- List files and directories
- Examine inode information
- Identify files associated with specific inode numbers
- Analyze deleted file information
- Recover files from the forensic image

Important TSK Commands Used

- **mmls** — Used to examine the partition layout.
- **fsstat** — Used to obtain detailed file-system information.
- **img_stat** — Used to examine information about the forensic image.
- **istat** — Used to examine metadata and MFT entries.
- **fls** — Used to list files and directories.

- **ils** — Used to display inode information.
- **ffind** — Used to identify a file associated with a specific inode.
- **tsk_recover** — Used to recover files from the forensic image.

Advantages of The Sleuth Kit

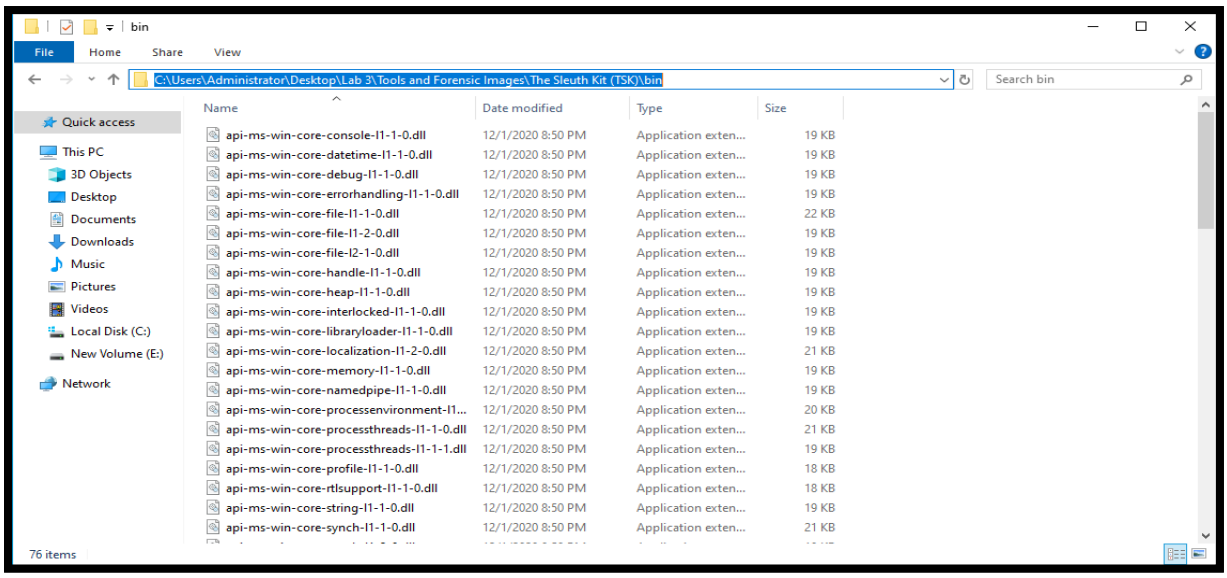
- Provides detailed low-level file-system analysis.
- Allows investigators to examine forensic evidence through command-line tools.
- Provides access to file-system metadata and internal structures.
- Supports analysis of deleted and existing files.
- Helps investigators examine NTFS and other supported file systems.
- Provides tools for file identification and recovery.

Windows File System Analysis Using The Sleuth Kit

Practical Demonstration

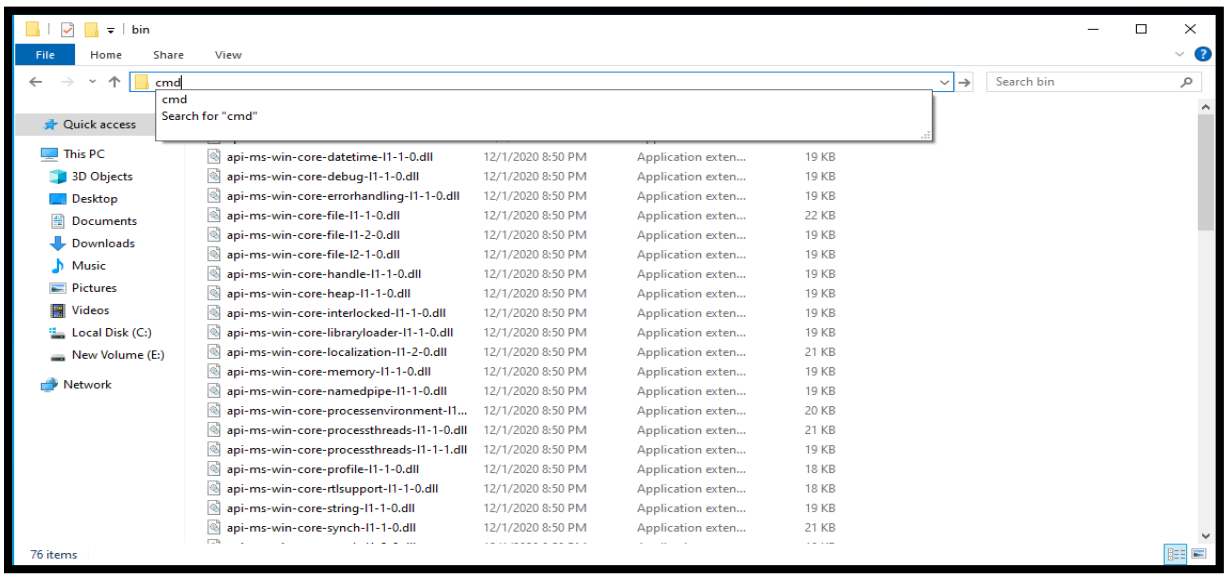
01. Opening The Sleuth Kit Directory

The Sleuth Kit tools were accessed from the Windows 10 machine through the provided TSK directory.



02. Opening Command Prompt

Command Prompt was opened to execute The Sleuth Kit forensic analysis commands.

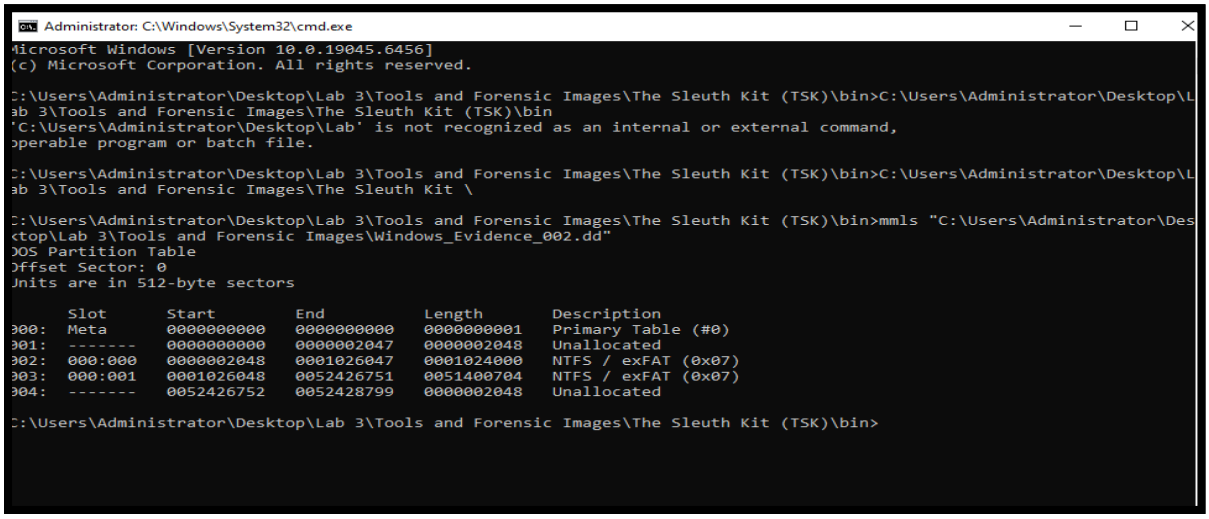


03. Partition Table Analysis Using mmls

The mmls command was used to examine the partition layout of the forensic image.

Purpose:

- Identify partition information
- Examine the volume system
- Understand the layout of the disk image

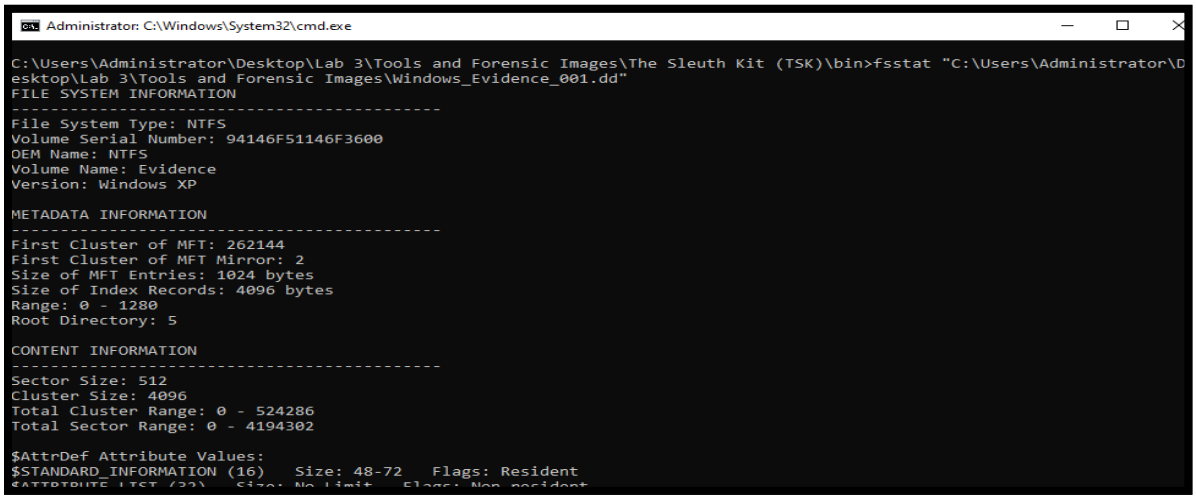


04. File System Analysis Using fsstat

The fsstat command was used to obtain information about the file system and operating system associated with the forensic image.

Result:

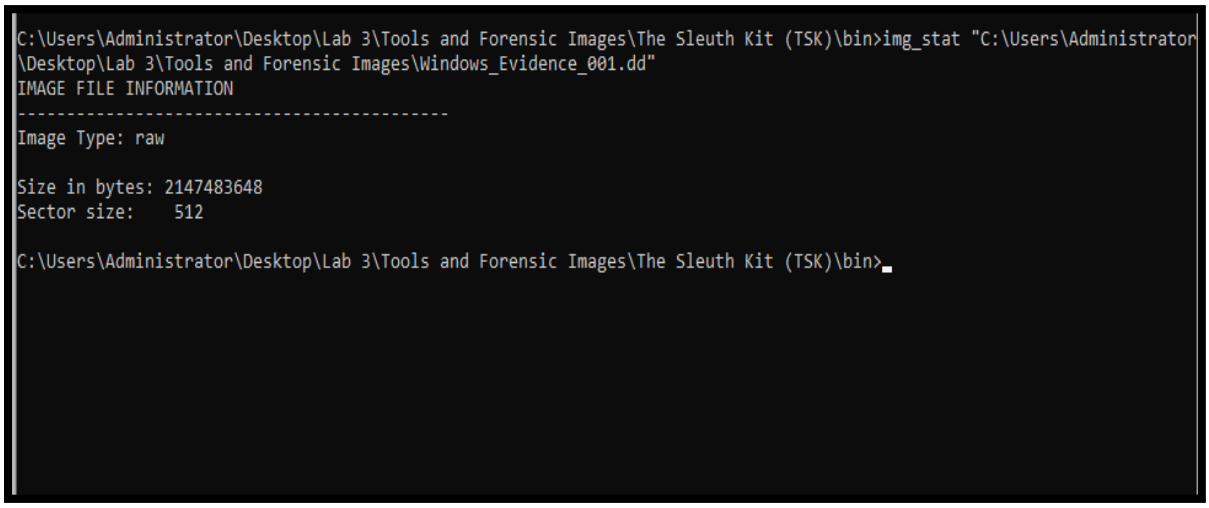
- File system identified as **NTFS**
- Source operating system identified as **Windows XP**



05. Image Analysis Using img_stat

The img_stat command was used to display information about the selected forensic image.

It provides details about the image that help investigators understand the structure and properties of the evidence file.



06. MFT Analysis Using istat

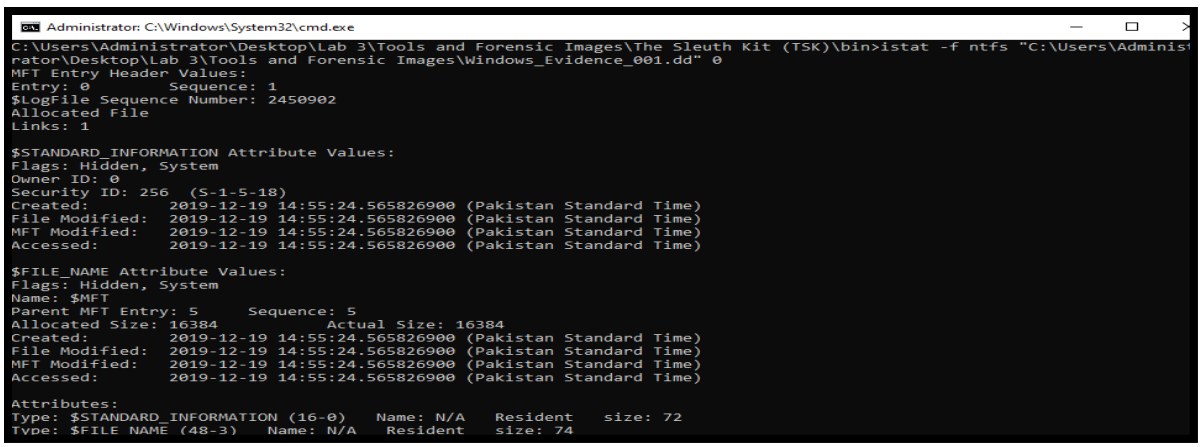
The istat command was used to examine the metadata structures of the NTFS file system.

The **Master File Table (MFT)** contains an entry for every file and directory in the NTFS file system.

MFT entry 0 was examined using the istat command.

Purpose:

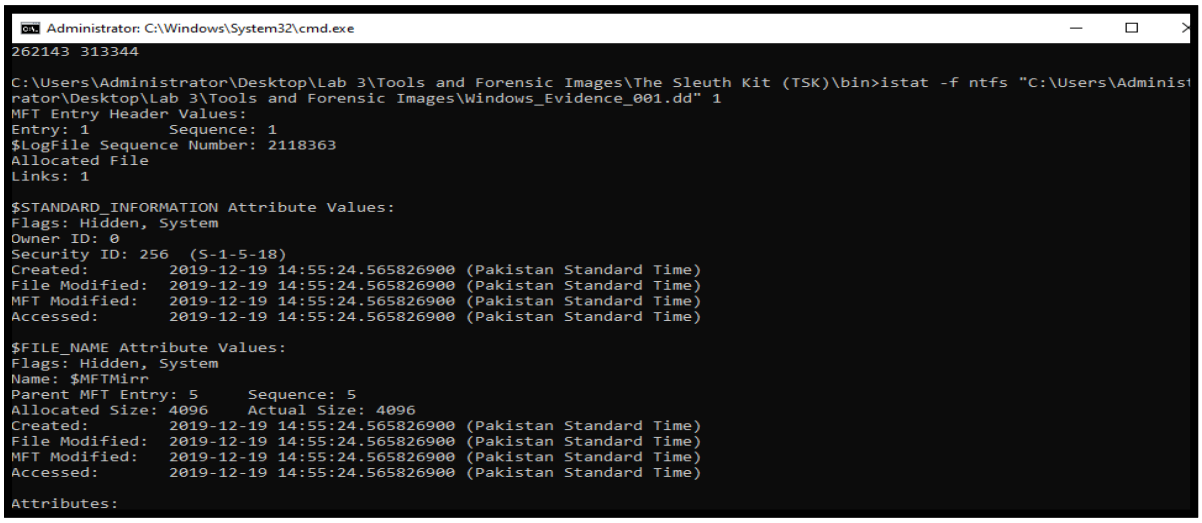
- View the MFT overview
- Understand the structure of the Master File Table
- Examine NTFS metadata information



07. MFTMirr Analysis

MFT entry 1 was examined to view the **MFTMirr** file.

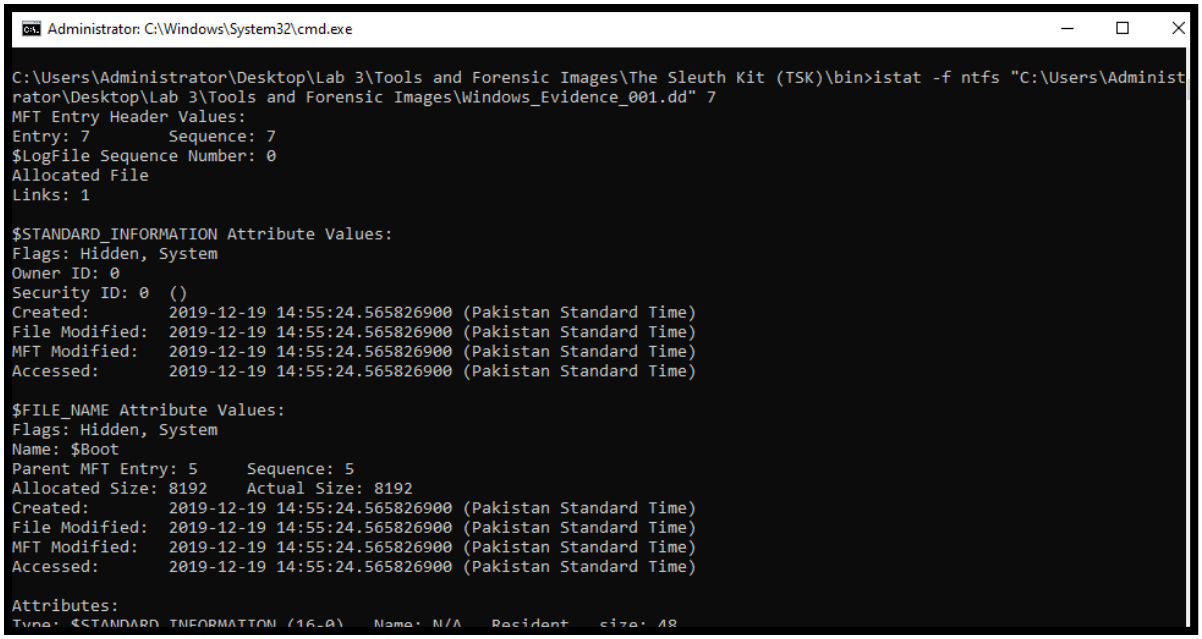
MFTMirr contains a backup copy of the first MFT entries and provides additional information about the NTFS metadata structure.



08. Boot File Analysis

MFT entry 7 was examined to view the Boot file.

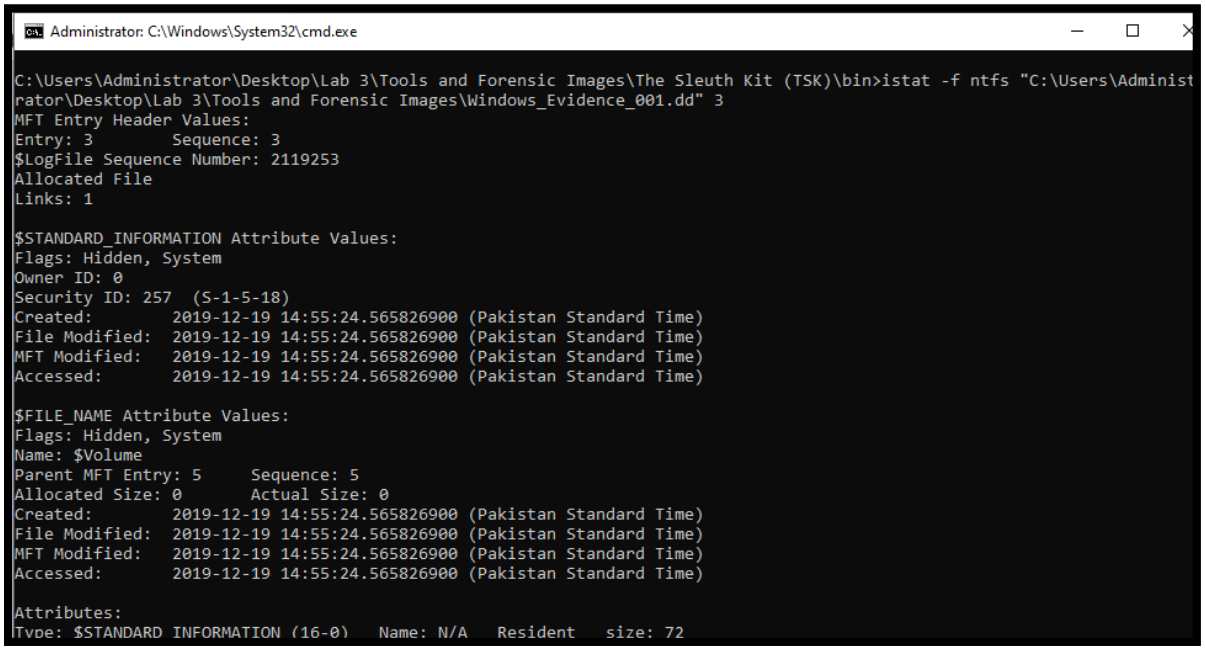
The Boot file contains the boot sector information associated with the NTFS file system.



09. Volume File Analysis

MFT entry 3 was examined to obtain information about the volume file system metadata.

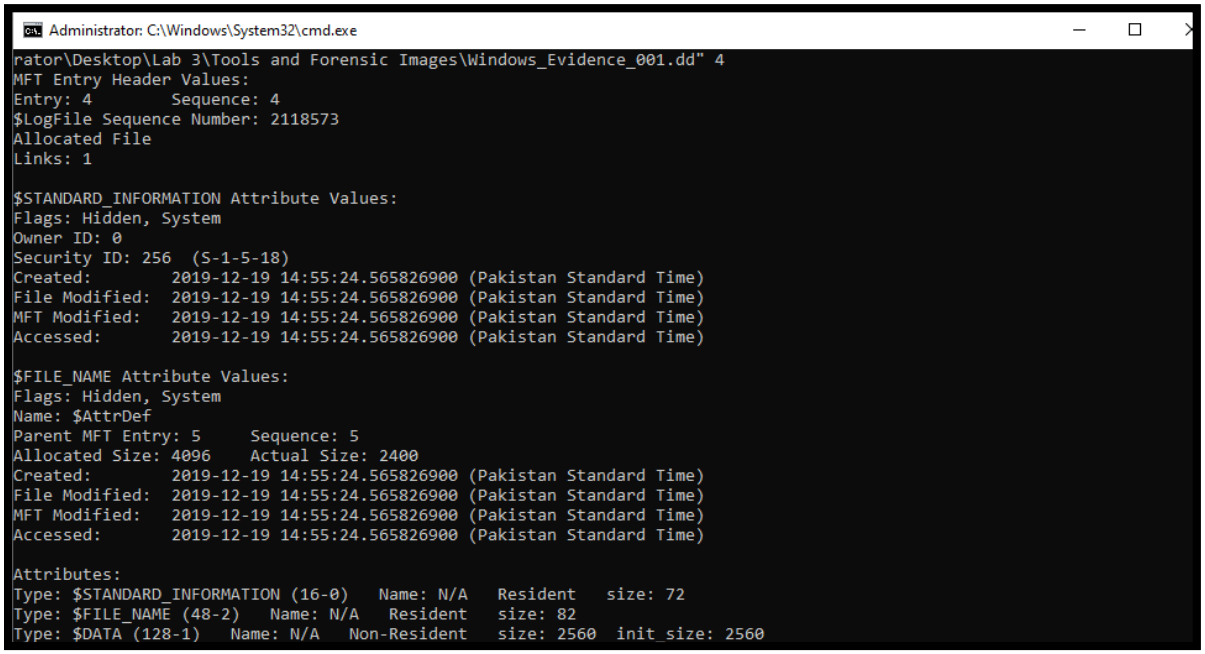
This includes information such as the volume label and version-related information.



10. AttrDef File Analysis

MFT entry 4 was examined to analyze the **AttrDef** file.

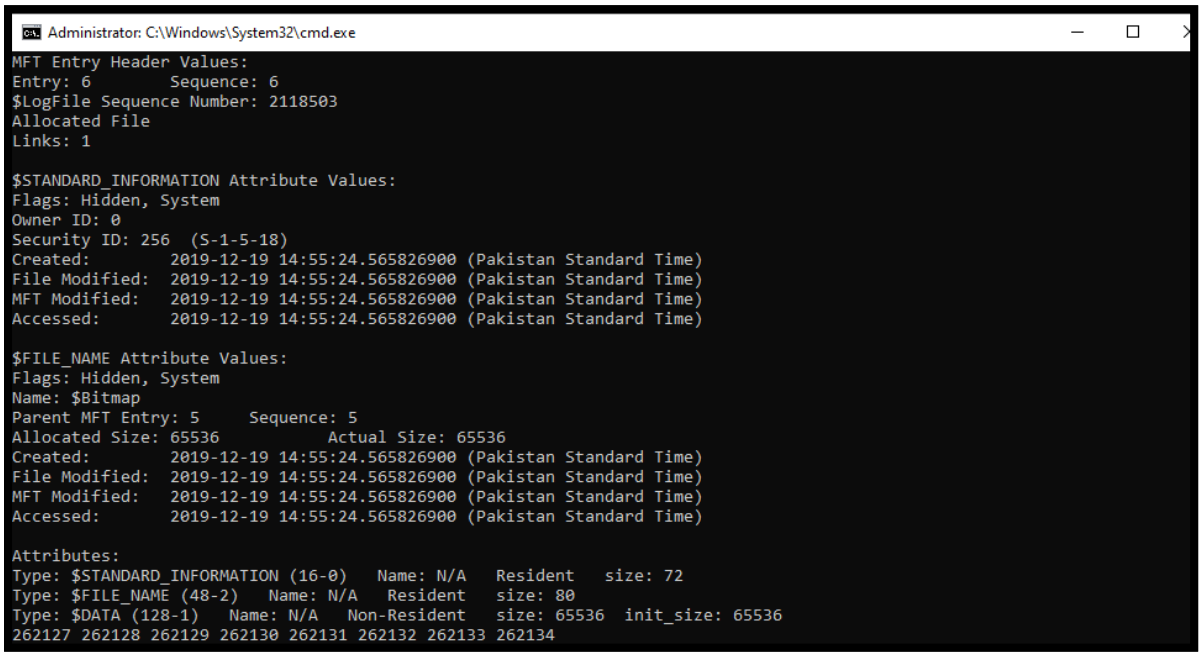
The AttrDef file defines the names and type identifiers for different NTFS attributes.



11. Bitmap File Analysis

MFT entry 6 was analyzed to examine the **Bitmap** file.

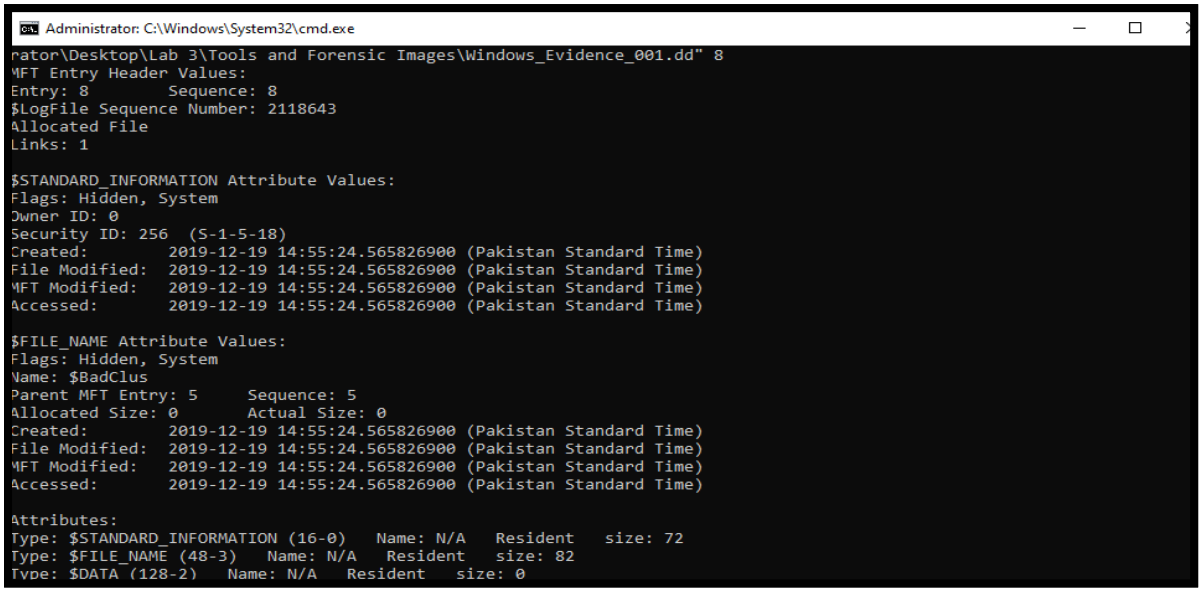
The Bitmap file is related to determining the allocation status of clusters within the NTFS file system.



12. BadClus File Analysis

MFT entry 8 was examined to analyze the **BadClus** file.

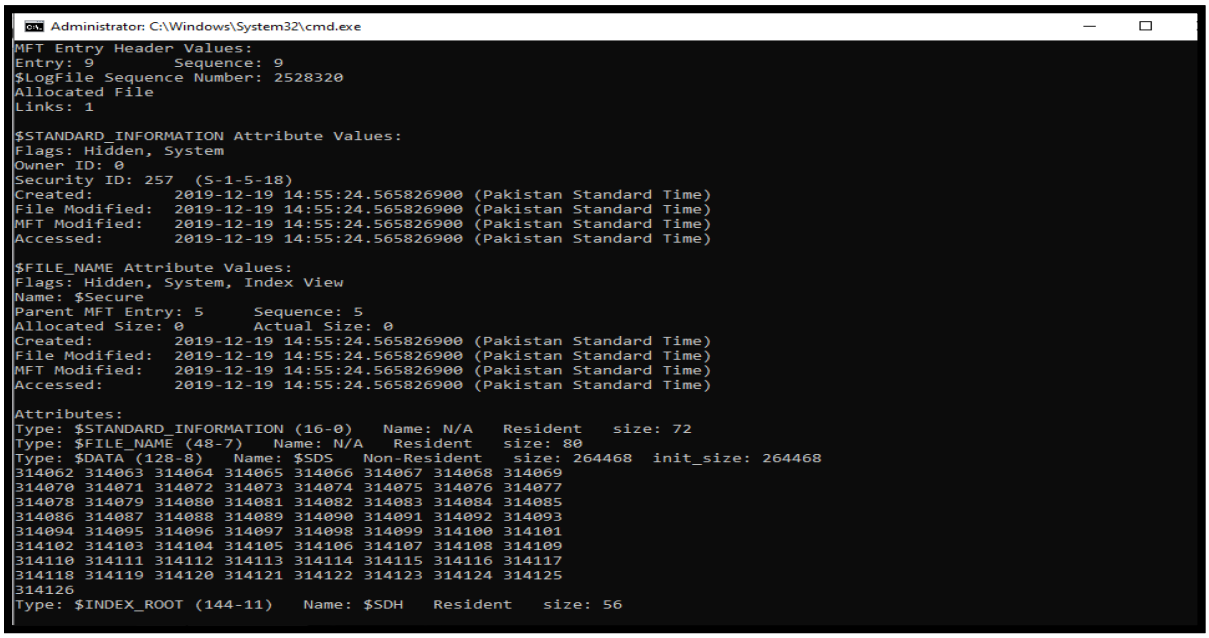
NTFS uses this metadata file to keep track of damaged or bad clusters.



13. Secure File Analysis

MFT entry 9 was examined to analyze the **Secure** file.

The Secure file contains security descriptors that define access control information for files and directories.

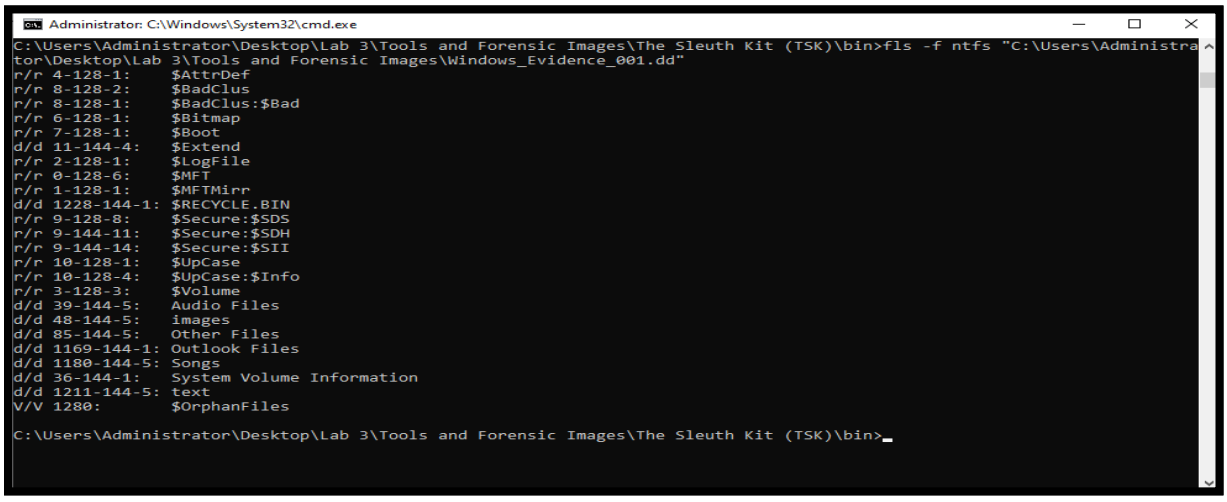


File and Directory Analysis

14. Listing Files Using fls

The fls command was used to list the files and directory names contained within the Windows forensic image.

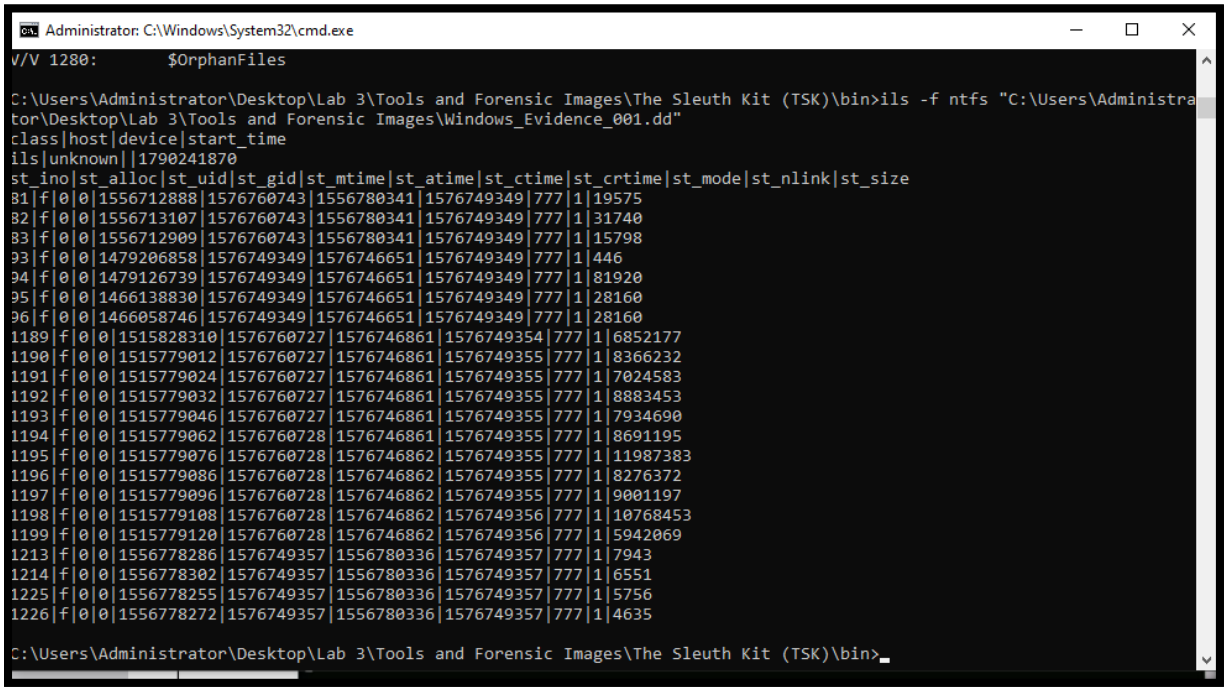
This helped identify the available file system contents.



15. Examining Inodes Using ils

The ils command was used to display inode information from the forensic image.

The -r option was used to display information related to removed files.

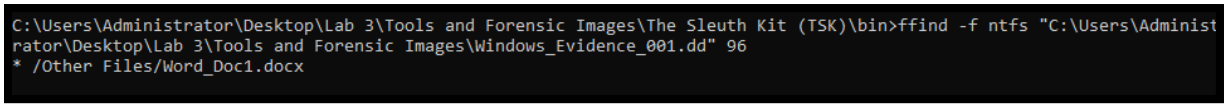


16. Finding a File Using ffind

The ffind command was used to identify the file or directory associated with a specific inode number.

Inode examined:

- Inode number: 96
- File identified: Word Doc1.docx



17. File Recovery Using tsk_recover

The tsk_recover tool was used to recover files from the forensic image.

Steps performed:

- Created a folder named Retrieved Files by SleuthKit
- Specified the Windows forensic image

- Used the raw image type
- Used the -e option to recover allocated and non-allocated files
- Stored the recovered files in the created folder

```
C:\Users\Administrator\Desktop\Lab 3\Tools and Forensic Images\The Sleuth Kit (TSK)\bin>tsk_recover -i raw -e "C:\Users\
Administrator\Desktop\Lab 3\Tools and Forensic Images\Windows_Evidence_001.dd" "C:\Users\Administrator\Desktop\Lab 3"
Files Recovered: 1097
```

18. Viewing Recovered Files

After the recovery process was completed, the recovered files and directories were opened from the **Retrieved Files by SleuthKit** folder.

This confirmed that files could be extracted from the forensic disk image for further examination.

